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TO: AGC4 Distribution  
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SUBJECT: AGC4 MEMO # 9 - Block II Instructions

## TABLE OF CONTENTS

|                                                       |    |
|-------------------------------------------------------|----|
| Introduction . . . . .                                | 2  |
| Memory . . . . .                                      | 3  |
| Basic Instructions . . . . .                          | 3  |
| Extracode Instructions . . . . .                      | 11 |
| Implied-Address Codes . . . . .                       | 15 |
| Unprogrammed Sequences . . . . .                      | 21 |
| Address Constant Formats . . . . .                    | 25 |
| Control Pulse Definitions . . . . .                   | 27 |
| Condensed List of Programmable Instructions . . . . . | 34 |
| Pulse Sequences . . . . .                             | 35 |

## Introduction

This document supercedes all revisions of and appendices to AGC4 Memo # 8, "Block II Instructions, Revised". The format has been changed to include more information for YUL-language programmers and to include the engineering details formerly relegated to appendices. A new descriptive section on unprogrammed sequences has been added.

Some confusion has arisen about the nature of channel numbers or addresses. Channel addresses should be used just like memory addresses in programming, that is, regarding the channels as a third category of memory, distinct from E and F. The fact that the numbers used as channel addresses coincide with some of the numbers used as memory addresses should cause no confusion, because the addresses in In/Out instructions are always channel addresses, and the addresses in other instructions are always memory addresses. In fact, the coincidence is put to good use: the L register is accessible both at memory address 0001 and at channel address 01.

In YUL language, symbols may be equated to channel addresses as well as memory addresses. The only distinction made by the assembler is that addresses of In/Out instructions have a theoretical maximum of 777.

## Memory

Block II differs significantly from Block I in register and memory layout and in addressing. The LP register has been renamed L because it is a lower accumulator in every sense. The IN and OUT registers no longer have addresses in memory, but are referenced with 9-bit channel addresses by the seven input/output instructions (code 10). Channel assignments are given in Digital Development Memo #254, Revision A (Sept. 7, 1965). Figures 1 and 2 show the arrangement of addresses. The erasable banks use local addresses 1400-1777. The fixed banks use local addresses 2000-3777. Figure 3 explains the bank-switching and editing registers.

## Basic Instructions

Figure 4 shows the relationships among the operation codes, with alternate spelling in brackets. Subscripts are running times, in MCT; EXTEND time of 1 MCT is not included in extracode times.

|                  |                                                                                                                                                                    |                  |       |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-------|
| Code 00.         | I: TC K                                                                                                                                                            | Transfer Control | 1 MCT |
| K $\neq$ 3, 4, 6 | Set $c(Q) = TC\ I + 1$ ;<br>Take next instruction from K and proceed from there.<br>Remarks: Alternate spelling is TCR, for Transfer<br>Control setting up Return. |                  |       |

# ARRANGEMENT OF ADDRESSES

| OCTAL PSEUDO-<br>ADDRESS | REGISTER<br>NAME                          | REMARKS                | TYPE                      |
|--------------------------|-------------------------------------------|------------------------|---------------------------|
| 00000                    | A                                         |                        | Flip-flop<br>registers    |
| 00001                    | L                                         | (also channel 01)      |                           |
| 00002                    | Q                                         | (also channel 02)      |                           |
| 00003                    | EB                                        | Erasable Bank Register |                           |
| 00004                    | FB                                        | Fixed Bank Register    |                           |
| 00005                    | Z                                         |                        |                           |
| 00006                    | BB                                        | Both Bank Registers    |                           |
| 00007                    | --                                        | Zeros                  |                           |
| 00010                    | ARUPT                                     | xRUPT = Storage for x  | 2040 words<br>of Erasable |
| 00011                    | LRUPT                                     | during Interrupt;      |                           |
| 00012                    | QRUPT                                     | ZRUPT & BRUPT stored   |                           |
| 00013                    | (spare)                                   | automatically.         |                           |
| 00014                    | (spare)                                   |                        |                           |
| 00015                    | ZRUPT                                     |                        |                           |
| 00016                    | BBRUPT                                    |                        |                           |
| 00017                    | BRUPT                                     | (RIP)                  |                           |
| 00020                    | CYR                                       | Cycle Right 1 Bit      |                           |
| 00021                    | SR                                        | Shift Right 1 Bit      |                           |
| 00022                    | CYL                                       | Cycle Left 1 Bit       |                           |
| 00023                    | EDOP                                      | Edit (Polish) Opcode   |                           |
| 00024-00057              | Counters                                  |                        |                           |
| 00060-01377              | Unswitched Erasable                       |                        |                           |
| 01400-03777              | 5 Erasable Banks @ 256 words (See Fig. 2) |                        |                           |
| 04000 -up                | Fixed (See Fig. 2)                        |                        | Fixed                     |

Fig. 1

# Fixed and Erasable Bank-Switching

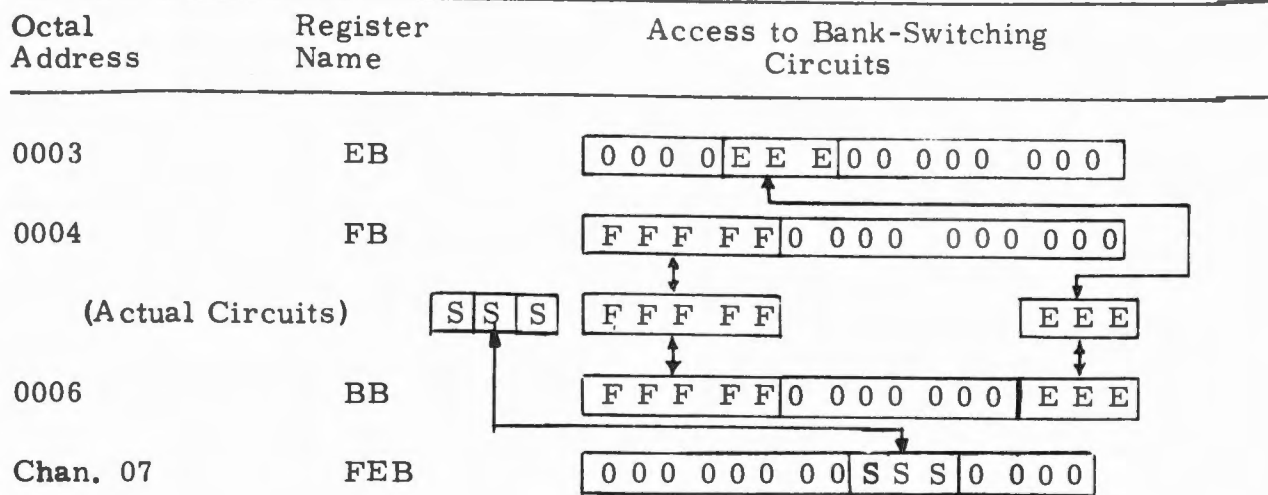
(Fig. 2)

| Octal Pseudo-Address  | Memory Type  | Erasable Bank Reg. | Fixed Bank Reg. | Fixed Ex-tension bits<br>(7, 6, 5 of channel 7) | S-Reg. Value |
|-----------------------|--------------|--------------------|-----------------|-------------------------------------------------|--------------|
| 00000-01377           | (Note 1)     | x                  | xx              | x                                               | 0000-1377    |
| 00000-00377           | (Note 1)     | 0                  | xx              | x                                               | 1400-1777    |
| 00400-00777           | Unswitched E | 1                  | xx              | x                                               | 1400-1777    |
| 01000-01377           | Unswitched E | 2                  | xx              | x                                               | 1400-1777    |
| 01400-01777           | Switched E   | 3                  | xx              | x                                               | 1400-1777    |
| 02000-02377           | Switched E   | 4                  | xx              | x                                               | 1400-1777    |
| 02400-02777           | Switched E   | 5                  | xx              | x                                               | 1400-1777    |
| 03000-03377           | Switched E   | 6                  | xx              | x                                               | 1400-1777    |
| 03400-03777           | Switched E   | 7                  | xx              | x                                               | 1400-1777    |
| 04000-07777           | Fixed-fixed  | x                  | xx              | x                                               | 4000-7777    |
| 10000-11777           | Common fixed | x                  | 00              | x                                               | 2000-3777    |
| 12000-13777           | Common fixed | x                  | 01              | x                                               | 2000-3777    |
| 04000-05777           | Fixed-fixed  | x                  | 02              | x                                               | 2000-3777    |
| 06000-07777           | Fixed-fixed  | x                  | 03              | x                                               | 2000-3777    |
| 20000-21777           | Common fixed | x                  | 04              | x                                               | 2000-3777    |
| 22000-23777           | Common fixed | x                  | 05              | x                                               | 2000-3777    |
| -- and so on through: |              |                    |                 |                                                 |              |
| 64000-65777           | Common fixed | x                  | 26              | x                                               | 2000-3777    |
| 66000-67777           | Common fixed | x                  | 27              | x                                               | 2000-3777    |
| 70000-71777           | Super-bank 3 | x                  | 30              | 3*                                              | 2000-3777    |
| 72000-73777           | Super-bank 3 | x                  | 31              | 3*                                              | 2000-3777    |
| -- and so on through: |              |                    |                 |                                                 |              |
| 106000-107777         | Super-bank 3 | x                  | 37              | 3*                                              | 2000-3777    |
| 110000-111777         | Super-bank 4 | x                  | 30              | 4                                               | 2000-3777    |
| 112000-113777         | Super-bank 4 | x                  | 31              | 4                                               | 2000-3777    |
| 114000-115777         | Super-bank 4 | x                  | 32              | 4                                               | 2000-3777    |
| 116000-117777         | Super-bank 4 | x                  | 33              | 4                                               | 2000-3777    |

(Note 1) Flip-flop central registers, counters, and unswitched erasable. Central and special-purpose registers will be accessed as E-bank 0 only under exceptional circumstances.

\* Any value 0-3 is OK. 3 is conventional.

# BANK-SWITCHING AND EDITING REGISTERS



A bank number written into EB or FB is automatically available at BB.

Information written into BB is automatically available at EB and FB.

## EDITING REGISTER TRANSFORMATIONS

| (bit positions) |      | 15 14 13 12 11 10 09 08 07 06 05 04 03 02 01 |
|-----------------|------|----------------------------------------------|
| 0020            | CYR  | 01 15 14 13 12 11 10 09 08 07 06 05 04 03 02 |
| 0021            | SR   | 15 15 14 13 12 11 10 09 08 07 06 05 04 03 02 |
| 0022            | CYL  | 14 13 12 11 10 09 08 07 06 05 04 03 02 01 15 |
| 0023            | EDOP | -- -- -- -- -- -- -- -- 14 13 12 11 10 09 08 |

Fig. 3

| 00                                                                                                                                                                                                                 | 01                                                      | 02                                                                                                            | 03                                                          | 04                         | 05                                                                                                                                            | 06                                                       | 07                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|---------------------------------------|
| <div>RELINT(3)</div> <div>INHINT(4)</div> <div>EXTEND(6)</div> <div>TC<sub>1</sub><br/>[TCR]</div>                                                                                                                 | <div>CCS<sub>2</sub></div> <div>TCF<sub>1</sub></div>   | <div>DAS<sub>3</sub></div> <div>LXCH<sub>2</sub></div> <div>INCR<sub>2</sub></div> <div>ADS<sub>2</sub></div> | <div>CA<sub>2</sub></div> <div>[CAF]</div> <div>[CAE]</div> | <div>CS<sub>2</sub></div>  | <div>RESUME(17)</div> <div>INDEX<sub>2</sub><br/>[NDX]</div> <div>DXCH<sub>3</sub></div> <div>TS<sub>2</sub></div> <div>XCH<sub>2</sub></div> | <div>AD<sub>2</sub></div>                                | <div>MASK<sub>2</sub><br/>[MSK]</div> |
| <div>READ<sub>2</sub></div> <div>WRITE<sub>2</sub></div> <div>RAND<sub>2</sub></div> <div>WAND<sub>2</sub></div> <div>ROR<sub>2</sub></div> <div>WOR<sub>2</sub></div> <div>RXOR<sub>2</sub></div> <div>----</div> | <div>DV<sub>6</sub></div> <div>BZF<sub>1, 2</sub></div> | <div>MSU<sub>2</sub></div> <div>QXCH<sub>2</sub></div> <div>AUG<sub>2</sub></div> <div>DIM<sub>2</sub></div>  | <div>DCA<sub>3</sub></div>                                  | <div>DCS<sub>3</sub></div> | <div>INDEX<sub>2</sub><br/>[ NDX]</div>                                                                                                       | <div>SU<sub>2</sub></div> <div>BZMF<sub>1, 2</sub></div> | <div>MP<sub>3</sub></div>             |
| 10                                                                                                                                                                                                                 | 11                                                      | 12                                                                                                            | 13                                                          | 14                         | 15                                                                                                                                            | 16                                                       | 17                                    |

OPERATION CODES (10-17 are extracodes)

Fig. 4



Code 00. I: TC K (Special Cases of TC) 1 MCT  
 K = 3, 4, or 6 Set indicator specified by K;  
 Take next instruction from I + 1.  
 Remarks: TC 3 = RELINT (allow interrupt),  
 TC 4 = INHINT (inhibit interrupt),  
 TC 6 = EXTEND (set extracode switch).

The extracode switch causes the next instruction to be an extracode. Any extracode except INDEX resets the switch. Interrupt is inhibited while the switch is on.

Code 01. I: CCS K Count, Compare and Skip 2 MCT  
 QC0 Set c(A) = DABS [b(K)];  
 Set c(K) = b(K), re-editing if K is 0020-0023.  
 Take next instruction from I + 1 if b(K) > + 0;  
 from I + 2 if b(K) = + 0;  
 from I + 3 if b(K) < - 0;  
 from I + 4 if b(K) = - 0.  
 Remarks: The Diminished Absolute Value of an integer x

is:

$$DABS(x) = \begin{cases} |x| - 1 & \text{if } |x| > 1 \\ + 0 & \text{if } |x| \leq 1 \end{cases}$$

Code 01. I: TCF K Transfer Control to Fixed 1 MCT  
 QC1-3 Take next instruction from K and proceed from there.  
 Remarks: QC n denotes Quarter Code n, where n is bits 12 and 11 of the instruction word.

Code 02. I: DAS K Double Add to Storage 3 MCT  
 QC 0 Set c(K, K+1) = b(A, L) + b(K, K+1), re-editing if K or K + 1 is 0020-0023;  
 If K ≠ 0, Set c(L) = + 0 and set c(A) = net overflow;  
 Take next instruction from I + 1.  
 Remarks: If positive (negative) overflow resulted from the double precision addition as a whole, the net overflow is + 1(-1), otherwise it is + 0. Notice that DAS A doubles the contents of the double precision accumulator — implied address code DDOUBL assembles as DAS A. Since the

hardware must operate on the low-order operands first, consider DAS as the operation code 20001, to which the address K is added to form the instruction.

Code 02. I: LXCH K Exchange L and K 2 MCT

QC1

Set  $c(L) = b'(K)$ ;

Set  $c(K) = b(L)$ , re-editing if K is 0020-0023;

Take next instruction from  $I + 1$ .

Remarks: The prime indicates overflow correction.

Code 02. I: INCR K Increment 2 MCT

QC2

Set  $c(K) = b(K) + 1$ , re-editing if K is 0020-0023;

Take next instruction from  $I + 1$ .

Remarks: INCR and two other codes, AUG and DIM, are slightly modified counter-increment sequences. Accordingly, if one of this group overflows when addressing a counter for which overflow during involuntary incrementing is supposed to cause an interrupt, the interrupt will happen. This is true also for chain-reaction increments like  $T_2$ , which is incremented after an overflow of  $T_1$ . It should be noted that all these three instructions, unlike the increment sequences, always operate in ones complement, even when addressing CDU counters.

Code 02. I: ADS K Add to storage 2 MCT

QC3

Set  $c(A)$ ,  $c(K) = b(K) + b(A)$ , re-editing if  $K = 0020-0023$ ;

Take next instruction from  $I + 1$ .

Code 03. I: CA K Clear and Add 2 MCT

Set  $c(A) = b(K)$ ;

Set  $c(K) = b(K)$ , re-editing if K is 0020-0023;

Take next instruction from  $I + 1$ .

Remarks: Alternate spelling CAF is permitted when referring to fixed memory; alternate spelling CAE is permitted when referring to erasable memory.

Code 04. I: CS K Clear and Subtract 2 MCT

Set  $c(A) = -b(K)$ ;

Set  $c(K) = b(K)$ , re-editing if K is 0020-0023;

Take next instruction from  $I + 1$ .

Code 05. I: INDEX K Index Next Instruction 2 MCT  
 QC0 Set  $c(K) = b(K)$ , re-editing if K is 0020-0023;  
 K  $\neq$  0017 Use  $[b(K) + c(I+1)]$  as the next instruction.  
 Remarks: The prime indicates overflow correction.

Code 05. I: INDEX 0017 Resume Interrupted Program 2 MCT  
 QC0 Set  $c(Z) = c(0015)$   
 K = 0017 Use  $c(0017)$  as the next instruction.  
 Remarks: The implied-address code RESUME assembles as  
 INDEX 17.

Code 05. I: DXCH K Double Exchange 3 MCT  
 QC1 Set  $c(A, L) = b(K, K+1)$ ;  
 Set  $c(K, K+1) = b(A, L)$ , re-editing if K or K + 1 is 0020-0023;  
 Take next instruction from I + 1.  
 Remarks: The final  $c(L)$  will be overflow -corrected. The  
 operation code should be treated as 52001 (see DAS, page 8).

The implied-address codes DTCF (DXCH FB) and DTCB (DXCH Z) are recognized. The idea is that a DXCH, by changing both Z and one of the bank registers, can be a "double-precision transfer control" that can jump banks and leave a D. P. return address in A and L.

Code 05. I: TS K Transfer to Storage 2 MCT  
 QC2 Set  $c(K) = b(A)$ , re-editing if K is 0020-0023;  
 If  $\pm$  overflow in  $b(A)$ , set  $c(A) = \pm 1$  and take next instruction  
 from I + 2;  
 If no overflow in  $b(A)$ , take next instruction from I + 1.  
 Remarks: TS A guarantees  $c(A) = b(A)$  but skips to I + 2 on  
 overflow. Implied-address code = OVSK.

Code 05. I: XCH K Exchange A and K 2 MCT  
 QC3 Set  $c(A) = b(K)$ ;  
 Set  $c(K) = b(A)$ , re-editing if K is 0020-0023;  
 Take next instruction from I + 1.

Code 06. I: AD K ADD 2 MCT  
 Set  $c(A) = b(A) + b(K)$ ;  
 Set  $c(K) = b(K)$ , re-editing if K is 0020-0023;  
 Take next instruction from I + 1.  
 Remarks: The OVCTR of Block I has been dropped.

**Code 07.**

I: MASK    K    Mask A by K

2 MCT

Set  $c(A) = b(A) \wedge c(K)$ :

Take next instruction from  $I + 1$ .

Remarks:  $\wedge$  denotes Boolean AND. Truth table for each bit

$$A \quad K \quad A \wedge K$$

0 0 0

0 1 0

1 0 0

1 1 1

MASK specifically omits to re-edit an argument from 0020-0023, in order to aid the interpreter and other software.

## Extracode Instructions

**Code 10.**

| I: READ | KC  | Read Channel KC |
|---------|-----|-----------------|
| 1       | 1   | 1               |
| 2       | 2   | 2               |
| 3       | 3   | 3               |
| 4       | 4   | 4               |
| 5       | 5   | 5               |
| 6       | 6   | 6               |
| 7       | 7   | 7               |
| 8       | 8   | 8               |
| 9       | 9   | 9               |
| 10      | 10  | 10              |
| 11      | 11  | 11              |
| 12      | 12  | 12              |
| 13      | 13  | 13              |
| 14      | 14  | 14              |
| 15      | 15  | 15              |
| 16      | 16  | 16              |
| 17      | 17  | 17              |
| 18      | 18  | 18              |
| 19      | 19  | 19              |
| 20      | 20  | 20              |
| 21      | 21  | 21              |
| 22      | 22  | 22              |
| 23      | 23  | 23              |
| 24      | 24  | 24              |
| 25      | 25  | 25              |
| 26      | 26  | 26              |
| 27      | 27  | 27              |
| 28      | 28  | 28              |
| 29      | 29  | 29              |
| 30      | 30  | 30              |
| 31      | 31  | 31              |
| 32      | 32  | 32              |
| 33      | 33  | 33              |
| 34      | 34  | 34              |
| 35      | 35  | 35              |
| 36      | 36  | 36              |
| 37      | 37  | 37              |
| 38      | 38  | 38              |
| 39      | 39  | 39              |
| 40      | 40  | 40              |
| 41      | 41  | 41              |
| 42      | 42  | 42              |
| 43      | 43  | 43              |
| 44      | 44  | 44              |
| 45      | 45  | 45              |
| 46      | 46  | 46              |
| 47      | 47  | 47              |
| 48      | 48  | 48              |
| 49      | 49  | 49              |
| 50      | 50  | 50              |
| 51      | 51  | 51              |
| 52      | 52  | 52              |
| 53      | 53  | 53              |
| 54      | 54  | 54              |
| 55      | 55  | 55              |
| 56      | 56  | 56              |
| 57      | 57  | 57              |
| 58      | 58  | 58              |
| 59      | 59  | 59              |
| 60      | 60  | 60              |
| 61      | 61  | 61              |
| 62      | 62  | 62              |
| 63      | 63  | 63              |
| 64      | 64  | 64              |
| 65      | 65  | 65              |
| 66      | 66  | 66              |
| 67      | 67  | 67              |
| 68      | 68  | 68              |
| 69      | 69  | 69              |
| 70      | 70  | 70              |
| 71      | 71  | 71              |
| 72      | 72  | 72              |
| 73      | 73  | 73              |
| 74      | 74  | 74              |
| 75      | 75  | 75              |
| 76      | 76  | 76              |
| 77      | 77  | 77              |
| 78      | 78  | 78              |
| 79      | 79  | 79              |
| 80      | 80  | 80              |
| 81      | 81  | 81              |
| 82      | 82  | 82              |
| 83      | 83  | 83              |
| 84      | 84  | 84              |
| 85      | 85  | 85              |
| 86      | 86  | 86              |
| 87      | 87  | 87              |
| 88      | 88  | 88              |
| 89      | 89  | 89              |
| 90      | 90  | 90              |
| 91      | 91  | 91              |
| 92      | 92  | 92              |
| 93      | 93  | 93              |
| 94      | 94  | 94              |
| 95      | 95  | 95              |
| 96      | 96  | 96              |
| 97      | 97  | 97              |
| 98      | 98  | 98              |
| 99      | 99  | 99              |
| 100     | 100 | 100             |

2 MCT

PCO

Set  $c(A) = c(KC)$ , where  $KC$  is an in/out channel:

Take next instruction from  $I + 1$ .

Remarks: Code 10 is broken down into seven peripheral codes

**Code 10.**

**I: WRITE KC      Write Channel KC**

2 MCT

PC1

Set  $c(KC) = c(A)$ :

Take next instruction from  $I + 1$ .

**Code 10.**

| I: RAND | KC  | Read and Mask |
|---------|-----|---------------|
| 0       | 0   | 0             |
| 0       | 1   | 0             |
| 0       | 2   | 0             |
| 0       | 3   | 0             |
| 0       | 4   | 0             |
| 0       | 5   | 0             |
| 0       | 6   | 0             |
| 0       | 7   | 0             |
| 0       | 8   | 0             |
| 0       | 9   | 0             |
| 0       | 10  | 0             |
| 0       | 11  | 0             |
| 0       | 12  | 0             |
| 0       | 13  | 0             |
| 0       | 14  | 0             |
| 0       | 15  | 0             |
| 0       | 16  | 0             |
| 0       | 17  | 0             |
| 0       | 18  | 0             |
| 0       | 19  | 0             |
| 0       | 20  | 0             |
| 0       | 21  | 0             |
| 0       | 22  | 0             |
| 0       | 23  | 0             |
| 0       | 24  | 0             |
| 0       | 25  | 0             |
| 0       | 26  | 0             |
| 0       | 27  | 0             |
| 0       | 28  | 0             |
| 0       | 29  | 0             |
| 0       | 30  | 0             |
| 0       | 31  | 0             |
| 0       | 32  | 0             |
| 0       | 33  | 0             |
| 0       | 34  | 0             |
| 0       | 35  | 0             |
| 0       | 36  | 0             |
| 0       | 37  | 0             |
| 0       | 38  | 0             |
| 0       | 39  | 0             |
| 0       | 40  | 0             |
| 0       | 41  | 0             |
| 0       | 42  | 0             |
| 0       | 43  | 0             |
| 0       | 44  | 0             |
| 0       | 45  | 0             |
| 0       | 46  | 0             |
| 0       | 47  | 0             |
| 0       | 48  | 0             |
| 0       | 49  | 0             |
| 0       | 50  | 0             |
| 0       | 51  | 0             |
| 0       | 52  | 0             |
| 0       | 53  | 0             |
| 0       | 54  | 0             |
| 0       | 55  | 0             |
| 0       | 56  | 0             |
| 0       | 57  | 0             |
| 0       | 58  | 0             |
| 0       | 59  | 0             |
| 0       | 60  | 0             |
| 0       | 61  | 0             |
| 0       | 62  | 0             |
| 0       | 63  | 0             |
| 0       | 64  | 0             |
| 0       | 65  | 0             |
| 0       | 66  | 0             |
| 0       | 67  | 0             |
| 0       | 68  | 0             |
| 0       | 69  | 0             |
| 0       | 70  | 0             |
| 0       | 71  | 0             |
| 0       | 72  | 0             |
| 0       | 73  | 0             |
| 0       | 74  | 0             |
| 0       | 75  | 0             |
| 0       | 76  | 0             |
| 0       | 77  | 0             |
| 0       | 78  | 0             |
| 0       | 79  | 0             |
| 0       | 80  | 0             |
| 0       | 81  | 0             |
| 0       | 82  | 0             |
| 0       | 83  | 0             |
| 0       | 84  | 0             |
| 0       | 85  | 0             |
| 0       | 86  | 0             |
| 0       | 87  | 0             |
| 0       | 88  | 0             |
| 0       | 89  | 0             |
| 0       | 90  | 0             |
| 0       | 91  | 0             |
| 0       | 92  | 0             |
| 0       | 93  | 0             |
| 0       | 94  | 0             |
| 0       | 95  | 0             |
| 0       | 96  | 0             |
| 0       | 97  | 0             |
| 0       | 98  | 0             |
| 0       | 99  | 0             |
| 0       | 100 | 0             |
| 0       | 101 | 0             |
| 0       | 102 | 0             |
| 0       | 103 | 0             |
| 0       | 104 | 0             |
| 0       | 105 | 0             |
| 0       | 106 | 0             |
| 0       | 107 | 0             |
| 0       | 108 | 0             |
| 0       | 109 | 0             |
| 0       | 110 | 0             |
| 0       | 111 | 0             |
| 0       | 112 | 0             |
| 0       | 113 | 0             |
| 0       | 114 | 0             |
| 0       | 115 | 0             |
| 0       | 116 | 0             |
| 0       | 117 | 0             |
| 0       | 118 | 0             |
| 0       | 119 | 0             |
| 0       | 120 | 0             |
| 0       | 121 | 0             |
| 0       | 122 | 0             |
| 0       | 123 | 0             |
| 0       | 124 | 0             |
| 0       | 125 | 0             |
| 0       | 126 | 0             |
| 0       | 127 | 0             |
| 0       | 128 | 0             |
| 0       | 129 | 0             |
| 0       | 130 | 0             |
| 0       | 131 | 0             |
| 0       | 132 | 0             |
| 0       | 133 | 0             |
| 0       | 134 | 0             |
| 0       | 135 | 0             |
| 0       | 136 | 0             |
| 0       | 137 | 0             |
| 0       | 138 |               |

2 MCT

PC2

Set  $c(A) = b(A) \wedge c(KC)$ ;

Take next instruction from  $I + 1$ .

Remarks:  $\wedge$  denotes Boolean AND (see MASK).

**Code 10.**

**I: WAND      KC      Write and Mask**

2 MCT

PC3

Set  $c(KC)$ ,  $c(A) = b(A) \wedge b(KC)$ ;

Take next instruction from  $I + 1$ .

**Code 10.**

**I: ROR KC      Read and Superimpose**

2 MCT

PC4

Set  $c(A) = b(A) \vee c(KC)$ ;

Take next instruction from I + 1.

Remarks:  $\vee$  denotes Boolean Inclusive OR. Truth table for each bit position of b(A) and c(KC):

| A | KC | $A \vee KC$ |
|---|----|-------------|
| 0 | 0  | 0           |
| 0 | 1  | 1           |
| 1 | 0  | 1           |
| 1 | 1  | 1           |

Code 10. I: WOR KC Write and Superimpose 2 MCT  
PC5 Set c(KC),  $c(A) = b(A) \vee b(KC)$ ;  
Take next instruction from I + 1.

Code 10. I: RXOR KC Read and Invert 2 MCT  
PC6 Set  $c(A) = b(A) \nabla c(KC)$ ;  
Take next instruction from I + 1.

Remarks:  $\nabla$  denotes Boolean Exclusive OR. Truth table for each bit position of b(A) and c(KC):

| A | KC | $A \nabla KC$ |
|---|----|---------------|
| 0 | 0  | 0             |
| 0 | 1  | 1             |
| 1 | 0  | 1             |
| 1 | 1  | 0             |

Code 10. EDRUPT 3 MCT  
PC7 (For machine checkout only)

Code 11. I: DV K Divide 6 MCT  
QC0 Set  $c(A) = b(A, L) \div c(K)$ ;  
Set  $c(L) = \text{remainder}$ ;  
Take next instruction from I + 1.

Remarks: The signs of the double-length dividend in A and L need not agree. The net sign of the dividend is the sign of b(A) unless  $b(A) = \pm 0$ , in which case it is the sign of b(L). The remainder bears the net dividend sign, and the quotient sign is determined strictly by the divisor and net dividend signs. DV does not disturb c(Q), and does not re-edit an argument from 0020-0023 because there isn't enough time.

Code 11            I: BZF K            Branch Zero to Fixed            1 or 2 MCT  
 QC 1-3            If  $C(A) = \pm 0$ , take next instruction from K and proceed from  
                          there (1 MCT);  
                          Otherwise, take next instruction from I + 1 (2 MCT).

Code 12.            I: MSU K            Modular Subtract            2 MCT  
 QC0                Set  $c(A) = b(A) \oplus b(K)$ ;  
                          Set  $c(K) = b(K)$ , re-editing if K is 0020-0023;  
                          Take next instruction from I + 1.  
                          Remarks:  $\oplus$  denotes modular subtraction, which forms a  
                          signed one's complement difference of two unsigned (modular, or periodic)  
                          two's complement inputs. The method is to form the two's complement differ-  
                          ence, to decrement it if it is negative, and to take the overflow-uncorrected  
                          sum as the result.

Code 12.            I: QXCH K            Exchange Q and K            2 MCT  
 QC1                Set  $c(Q) = b(K)$ ;  
                          Set  $c(K) = b(Q)$ , re-editing if K is 002-0023;  
                          Take next instruction from I + 1.

Code 12.            I: AUG K            Augment            2 MCT  
 QC2                If  $b(K) \geq +0$ , set  $c(K) = b(K) + 1$ , re-editing if K is 0020-0023;  
                          If  $b(K) \leq -0$ , set  $c(K) = b(K) - 1$ , re-editing if K is 0020-0023;  
                          Take next instruction from I + 1.

Code 12.            I: DIM K            Diminish            2 MCT  
 QC3                If  $b(K) > +0$ , set  $c(K) = b(K) - 1$ , re-editing if K is 0020-0023;  
                          If  $b(K) = \pm 0$ , set  $c(K) = b(K)$ , re-editing if K is 0020-0023;  
                          If  $b(K) < -0$ , set  $c(K) = b(K) + 1$ , re-editing if K is 0020-0023;  
                          Take next instruction from I + 1.  
                          Remarks: DIM does not generate output pulses as DINC does.

Code 13.            I: DCA K            Double Clear and Add            3 MCT  
                          Set  $c(A, L) = b(K, K+1)$ ;  
                          Set  $c(K) = b(K)$ , re-editing if K is 0020-0023;  
                          Set  $c(K+1) = b(K+1)$ , re-editing if K+1 is 0020-0023;  
                          Take next instruction from I + 1.  
                          Remarks: The final  $c(L)$  will be overflow-corrected. The  
                          operation code should be treated as 30001 (see DAS, page 8).

Code 14. I: DCS K Double Clear and Subtract 3 MCT  
 Set  $c(A, L) = -b(K, K+1)$ ;  
 Set  $c(K) = b(K)$ , re-editing if K is 0020-0023;  
 Set  $c(K+1) = b(K+1)$ , re-editing if K+1 is 0020-0023;  
 Take next instruction from I + 1.

Remarks: DCS A succeeds in complementing the double precision accumulator — implied-address code: DCOM. The final  $c(L)$  will be overflow-corrected. The operation code should be treated as 40001 (see DAS page 8).

Code 15. I: INDEX K Index Extracode Instruction 2 MCT  
 (See INDEX, page 10).

Remarks: This is the only extracode that does not reset the extracode switch. The way to index an extracode (MP, say) is:

```
EXTEND
INDEX   ADDRWD
MP      0
```

The extension (extracode switch) will stay in force during any n-level nesting of extracode INDEXes. This INDEX will never act as a RESUME.

Code 16. I: SU K Subtract 2 MCT  
 QC0 Set  $c(A) = b(A) - b(K)$ ;  
 Set  $c(K) = b(K)$ , re-editing if K is 0020-0023;  
 Take next instruction from I + 1.

Code 16. I: BZMF K Branch Zero or Minus to Fixed 1 or 2 MCT  
 QC 1-3 If  $c(A) \leq +0$ , take next instruction from K and proceed from there (1 MCT);

Otherwise, take next instruction from I + 1 (2 MCT)

Code 17. I: MP K Multiply 3 MCT  
 Set  $c(A, L) = b(A) \times c(K)$ ;  
 Take next instruction from I + 1.

Remarks: The two words of the product agree in sign. A zero result is positive unless  $b(A) = \pm 0$  and  $c(K)$  is non-zero with the opposite sign.

MP does not re-edit an argument from 0020-0023 because there isn't enough time.

Some operations are defined for only one address value, like RESUME; others have unusual results when addressing central registers. For convenience in using these operation, the YUL System assembler recognizes implied-address codes, written without an address, and fills in the address. These codes are shown in Fig. 5 (alphabetically) and Fig. 6 (by actual code). Brief descriptions follow:

K = 0000

I: XXALQ      Execute Extracode

2 MCT

### Using A, L and Q

Assume that  $b(A) = 000006$  and  $b(L)$  is an extracode

instruction;

Execute the EXTEND in A, the instruction in L, then return to I + 1; leave c(Q) = 000003.

Remarks: This is a marginally useful operation because an extracode instruction built up in L could usually be executed better by the sequence:

## EXTEND

INDEX L

0 0

|          |        |                       |       |
|----------|--------|-----------------------|-------|
| Code 00. | I: XLQ | Execute using L and Q | 2 MCT |
|----------|--------|-----------------------|-------|

**K = 0001**

Assume that  $b(L)$  is a basic instruction.

Execute the instruction in L and, if it is not a successful branch, return to  $I + 1$ ;

Leave  $c(Q) = 000003$ .

Remarks: Like XXALQ, this operation is marginal. The time (2 MCT) for XXALQ and XLQ includes the TC to A or L and the return TC from Q, but not the time spent in executing c(A) or c(L).

|                 |                                         |              |
|-----------------|-----------------------------------------|--------------|
| <b>Code 00.</b> | <b>I: RETURN</b> Return from Subroutine | <b>2 MCT</b> |
|-----------------|-----------------------------------------|--------------|

**K = 0002**

**I: RETURN**    Return from Subroutine

2 MCT

Assume that  $b(Q) = TC = K'$ ;



from there;  
 Take the next instruction from K' and proceed

Leave  $c(Q) = 000003$ .

Code 00.            I: RELINT    Release (allow) Interrupt            1 MCT

K = 0003            Allow interrupt after this instruction (subject  
 to the restriction that interrupt cannot occur while there is  $\pm$  over-  
 flow in A);

Take next instruction from I + 1.

Code 00.            I: INHINT    Inhibit Interrupt            1 MCT

K = 0004            Inhibit interrupt until a subsequent RELINT;  
 Take next instruction from I + 1.

Remarks: The inhibition set by INHINT and removed  
 by RELINT is entirely independent of the one set by interrupt and  
 removed by RESUME.

Code 00.            I: EXTEND    Extend Next Instruction            1 MCT

K = 0006            Take the next instruction from I + 1 and execute  
 it as an extracode.

Remarks: If the next instruction is INDEX (full code 15),  
 the following instruction will be executed as an extracode too.

Code 01.            I: NOOP      No Operation (Fixed)            1 MCT

QC 1 - 3            Take the next instruction from I + 1.

K = I + 1            Remarks: This is how NOOP is assembled when I  
 is in fixed memory.

Code 02.            I: DDOUBL    Double Precision Double            3 MCT

QC 0                Set  $c(A, L) = b(A, L) + b(A, L)$ ;

K = 0000            Take next instruction from I + 1.

Remarks: If  $b(A)$  contains  $\pm$  overflow, the results  
 are messy; in particular,  $\text{sgn}[c(A)] \neq \text{sgn}[b(A)]$ . If  $|b(A)| \geq 1/2$ ,  
 overflow will be retained in  $c(A)$ .

# IMPLIED ADDRESS CODES

| Implied-Address Code | Actual Operation Code | Register (If applicable) | Word as assembled | NOTE |
|----------------------|-----------------------|--------------------------|-------------------|------|
| COM                  | CS                    | A                        | 40000             |      |
| DCOM                 | DCS                   | A                        | 40001             | X    |
| DDOUBL               | DAS                   | A                        | 20001             |      |
| DOUBLE               | AD                    | A                        | 60000             |      |
| DTCB                 | DXCH                  | Z                        | 52006             |      |
| <hr/>                |                       |                          |                   |      |
| DTCF                 | DXCH                  | FB                       | 52005             |      |
| EXTEND               | TC                    |                          | 00006             | S    |
| INHINT               | TC                    |                          | 00004             | S    |
| NOOP                 | TCF                   |                          | 1 (I+1)           | F    |
| NOOP                 | CA                    | A                        | 30000             | E    |
| <hr/>                |                       |                          |                   |      |
| OVSF                 | TS                    | A                        | 54000             |      |
| RELINT               | TC                    |                          | 00003             | S    |
| RESUME               | INDEX                 | BRUPT                    | 50017             | R    |
| RETURN               | TC                    | Q                        | 00002             |      |
| SQUARE               | MP                    | A                        | 70000             | X    |
| <hr/>                |                       |                          |                   |      |
| TCAA                 | TS                    | Z                        | 54005             |      |
| XLQ                  | TC                    | L                        | 00001             |      |
| XXALQ                | TC                    | A                        | 00000             |      |
| ZL                   | LXCH                  |                          | 22007             |      |
| ZQ                   | QXCH                  |                          | 22007             | X    |

## NOTE EXPLANATION:

|   |                                                                                |
|---|--------------------------------------------------------------------------------|
| E | Applies when I (location of instruction) is in erasable memory.                |
| F | Applies when I is in fixed memory.                                             |
| R | Special RESUME hardware responds to address 0017.                              |
| S | Special Indicator-setting hardware responds to addresses 0003, 0004, and 0006. |
| X | Extracode instruction.                                                         |

Fig. 5

IMPLIED ADDRESS CODES  
(By Actual Code)

| Actual<br>Operation<br>Code | Register<br>(If ap-<br>plicable) | Word as<br>assembled | Implied-<br>Address<br>Code | NOTE<br>(See<br>Fig. 5) |
|-----------------------------|----------------------------------|----------------------|-----------------------------|-------------------------|
| TC                          | A                                | 00000                | XXALQ                       |                         |
| TC                          | L                                | 00001                | XLQ                         |                         |
| TC                          | Q                                | 00002                | RETURN                      |                         |
| TC                          |                                  | 00003                | RELINT                      | S                       |
| TC                          |                                  | 00004                | INHINT                      | S                       |
| TC                          |                                  | 00006                | EXTEND                      | S                       |
| <hr/>                       |                                  |                      |                             |                         |
| TCF                         |                                  | 1 (I+1)              | NOOP                        | F                       |
| DAS                         | A                                | 20001                | DDOUBL                      |                         |
| LXCH                        |                                  | 22007                | ZL                          |                         |
| CA                          | A                                | 30000                | NOOP                        | E                       |
| CS                          | A                                | 40000                | COM                         |                         |
| INDEX                       | BRUPT                            | 50017                | RESUME                      | R                       |
| <hr/>                       |                                  |                      |                             |                         |
| DXCH                        | FB                               | 52005                | DTCF                        |                         |
| DXCH                        | Z                                | 52006                | DTCB                        |                         |
| TS                          | A                                | 54000                | OVSK                        |                         |
| TS                          | Z                                | 54005                | TCAA                        |                         |
| AD                          | A                                | 60000                | DOUBLE                      |                         |
| <hr/>                       |                                  |                      |                             |                         |
| QXCH                        |                                  | 22007                | ZQ                          | X                       |
| DCS                         | A                                | 40001                | DCOM                        | X                       |
| MP                          | A                                | 70000                | SQUARE                      | X                       |

Fig. 6

|          |                                                                                                                                                                                                                                                                        |                            |       |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-------|
| Code 02. | I: ZL                                                                                                                                                                                                                                                                  | Zero L                     | 2 MCT |
| QC 1     | Set $c(L) = +0$ ;                                                                                                                                                                                                                                                      |                            |       |
| K = 0007 | Take next instruction from $I + 1$ .                                                                                                                                                                                                                                   |                            |       |
|          | Remarks: This code and its companion ZQ depend on two properties of address 0007: no storage is associated with it, and references to it (in fact, to any of 0000-0007) are not checked for good parity. Address 0007 is therefore a generally usable source of zeros. |                            |       |
|          |                                                                                                                                                                                                                                                                        |                            |       |
| Code 03  | I: NOOP                                                                                                                                                                                                                                                                | No Operation (Erasable)    | 2 MCT |
| K = 0000 | Take next instruction from $I + 1$ .                                                                                                                                                                                                                                   |                            |       |
|          | Remarks: This is how NOOP is assembled when I is in erasable memory.                                                                                                                                                                                                   |                            |       |
|          |                                                                                                                                                                                                                                                                        |                            |       |
| Code 04. | I: COM                                                                                                                                                                                                                                                                 | Complement $c(A)$          | 2 MCT |
| K = 0000 | Set $c(A) = -b(A)$ ;                                                                                                                                                                                                                                                   |                            |       |
|          | Take next instruction from $I + 1$ .                                                                                                                                                                                                                                   |                            |       |
|          | Remarks: All 16 bits are complemented.                                                                                                                                                                                                                                 |                            |       |
|          |                                                                                                                                                                                                                                                                        |                            |       |
| Code 05. | I: RESUME                                                                                                                                                                                                                                                              | Resume Interrupted Program | 2 MCT |
| QC 0     | Set $c(Z) = c(0015)$ ;                                                                                                                                                                                                                                                 |                            |       |
| K = 0017 | Use $c(0017)$ as the next instruction.                                                                                                                                                                                                                                 |                            |       |
|          |                                                                                                                                                                                                                                                                        |                            |       |
| Code 05. | I: DTCF                                                                                                                                                                                                                                                                | Double Transfer Control,   | 3 MCT |
| QC 1     | Switching F bank                                                                                                                                                                                                                                                       |                            |       |
| K = 0004 | Set $c(A, L) = b(FB, Z)$ ;                                                                                                                                                                                                                                             |                            |       |
|          | Set $c(FB, Z) = b(A, L)$ ;                                                                                                                                                                                                                                             |                            |       |
|          | Take next instruction from new $c(Z)$                                                                                                                                                                                                                                  |                            |       |
|          | Remarks: A double-precision address constant format, 2 FCADR, is defined for use with DTCF.                                                                                                                                                                            |                            |       |

|          |                                                                                                                                                                                                                                                                |                         |       |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|-------|
| Code 05. | I: DTCB                                                                                                                                                                                                                                                        | Double Transfer Control | 3 MCT |
| QC 1     |                                                                                                                                                                                                                                                                | Switching Both Banks    |       |
| K = 0005 | Set $c(A, L) = b(Z, BB)$ ;<br>Set $c(Z, BB) = b(A, L)$ ;<br>Take next instruction from new $c(Z)$<br>Remarks: A double-precision address constant<br>format, 2 BCADR, is defined for use with DTCB.                                                            |                         |       |
|          |                                                                                                                                                                                                                                                                |                         |       |
| Code 05. | I: OVSK                                                                                                                                                                                                                                                        | Overflow Skip           | 2 MCT |
| QC 2     | Do not change $c(A)$ ;                                                                                                                                                                                                                                         |                         |       |
| K = 0000 | If <u>+</u> overflow in $c(A)$ , take next instruction from $I + 2$ ;<br>If no overflow in $c(A)$ , take next instruction from $I + 1$ .                                                                                                                       |                         |       |
|          |                                                                                                                                                                                                                                                                |                         |       |
| Code 05. | I: TCAA                                                                                                                                                                                                                                                        | Transfer Control to     | 2 MCT |
| QC 2     | Address in A                                                                                                                                                                                                                                                   |                         |       |
| K = 0005 | If <u>+</u> overflow in $b(A)$ , set $c(A) = \underline{+} 1$ ;<br>Take next instruction from the location whose address is<br>in bits 12-1 of $b(A)$ .<br>Remarks: The perils associated with TCAA in Mod<br>3C and Block I AGC do not exist in Block II AGC. |                         |       |
|          |                                                                                                                                                                                                                                                                |                         |       |
| Code 06. | I: DOUBLE                                                                                                                                                                                                                                                      | Double $c(A)$           | 2 MCT |
| K = 0000 | Set $c(A) = b(A) + b(A)$ ;<br>Take next instruction from $I + 1$ .<br>Remarks: See remarks on overflow under DDOUBL.                                                                                                                                           |                         |       |
|          |                                                                                                                                                                                                                                                                |                         |       |
| Code 12. | I: ZQ                                                                                                                                                                                                                                                          | Zero Q                  | 2 MCT |
| QC 1     | Set $c(Q) = + 0$ ;                                                                                                                                                                                                                                             |                         |       |
| K = 0007 | Take next instruction from $I + 1$ .<br>Remarks: See under ZL.                                                                                                                                                                                                 |                         |       |

|          |                                                                                                                                                   |                   |       |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------|
| Code 14. | I: DCOM                                                                                                                                           | Double Complement | 3 MCT |
| K = 0000 | Set $c(A, L) = -b(A, L)$ ;<br>Take next instruction from $I + 1$ .<br>Remarks: All 32 bits of A and L are complemented.                           |                   |       |
|          |                                                                                                                                                   |                   |       |
| Code 17. | I: SQUARE                                                                                                                                         | Square $c(A)$     | 3 MCT |
| K = 0000 | Set $c(A, L) = b(A) \times b(A)$ ,<br>Take next instruction from $I + 1$ .<br>Remarks: Results are messy if $b(A)$ contains <u>+</u><br>overflow. |                   |       |

### Unprogrammed Sequences

Some of the actions performed by the computer are not programmed but occur in response to external events. The categories of these unprogrammed sequences are shown in Fig. 7. Interrupt is inhibited if an interrupt has occurred after the latest RESUME, or an INHINT has occurred after the latest RELINT, or  $c(A)$  contains + overflow. Otherwise interrupt may occur before any basic (non-extracode) instruction except RELINT, INHINT, or EXTEND.

|      |                                                                                                                                                                                                                                                                                                           |       |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| RUPT | Interrupt Program                                                                                                                                                                                                                                                                                         | 3 MCT |
|      | Set $c(0015) = b(Z)$ ;<br>Set $c(0017) =$ the postponed instruction;<br>Take next instruction from the location whose address<br>is permanently associated with the cause of the interrupt, and proceed<br>from there. Inhibit further interrupt until RESUME.<br>Remarks: See also remarks under INHINT. |       |

Counter increments and decrements, serial-parallel conversion steps, and GSE interface transactions are lumped together under the name of counter interrupts because they perform limited tasks by snatching one or two memory cycles and then let the computer continue. They can occur before any instruction except RELINT, INHINT or EXTEND, but never inside instructions.

## UNPROGRAMMED SEQUENCES

|                                                |                                      |
|------------------------------------------------|--------------------------------------|
| Program Interrupt                              | RUPT                                 |
| Counter Increment/Decrement                    | PINC<br>PCDU<br>MINC<br>MCDU<br>DINC |
| Serial-Parallel Conversion<br>(and vice-versa) | SHINC<br>SHANC                       |
| Ground Support Interface                       | INOTRD<br>INOTLD<br>FETCH<br>STORE   |
| Manual Override                                | GOJ<br>TCSAJ                         |

Fig. 7

|      |                |       |
|------|----------------|-------|
| PINC | Plus Increment | 1 MCT |
|------|----------------|-------|

Set  $c(CTR) = b(CTR) + 1$ ;

If + overflow, set  $c(CTR) = + 0$  and set up an interrupt if  $CTR = T3, T4$  or  $T5$  or set up a PINC for  $T2$  if  $CTR = T1$ .

Remarks: This sequence and its priority chain effects are shared by the instruction INCR.

|      |                      |       |
|------|----------------------|-------|
| PCDU | Plus Increment (CDU) | 1 MCT |
|------|----------------------|-------|

Set  $c(CDUCTR) = b(CDUCTR) + 1$  in two's complement modular notation.

Remarks: Incrementing in two's-complement modular notation transforms 77777 into 00000 and 37777 into 40000, and is otherwise like one's-complement. INCR never acts like PCDU. PCDU and MCDU replace PINC and MINC for counters 0032-0036.

|      |                 |       |
|------|-----------------|-------|
| MINC | Minus Increment | 1 MCT |
|------|-----------------|-------|

Set  $c(CTR) = b(CTR) - 1$ ;

If - overflow, set  $c(CTR) = - 0$ .

|      |                      |       |
|------|----------------------|-------|
| MCDU | Minus Increment(CDU) | 1 MCT |
|------|----------------------|-------|

Set  $c(CDUCTR) = c(CDUCTR) - 1$  in twos complement modular notation.

Remarks: Transforms 40000 into 37777 and 00000 into 77777. See remarks under PCDU.

|      |                       |       |
|------|-----------------------|-------|
| DINC | Diminishing Increment | 1 MCT |
|------|-----------------------|-------|

If  $c(CTR) > + 0$ , set  $c(CTR) = b(CTR) - 1$  and emit signal POUT (Plus Output);

If  $c(CTR) < - 0$ , set  $c(CTR) = b(CTR) + 1$  and emit signal MOUT (Minus Output);

If  $c(CTR) = \pm 0$ , leave  $c(CTR)$  unchanged and emit signal ZOUT (Zero Output & turn off DINC request).



Remarks: Used to generate output pulse trains and to count down T6. Values to be counted down by DINC might be developed by the instruction MSU from a desired and an actual CDU angle. This sequence is shared by the instruction DIM, but without POUT, MOUT and ZOUT.

SHINC                      Shift Increment                      1 MCT

Set  $c(CTR) = b(CTR) + b(CTR)$ ;

If + overflow, set the priority chain station

for this counter.

Remarks: SHINC and SHANC are used to convert incoming serial bit streams into words for parallel access, and to convert words to outgoing serial bit streams.

SHANC                      Shift and Add Increment                      1 MCT

Set  $c(CTR) = b(CTR) + b(CTR) + 1$ ;

If + overflow, set the priority chain station

for this counter.

Remarks: See under SHINC.

INOTRD                      In/Out Read to GSE                      1 MCT

Accept a channel address from the Ground Support Equipment and place the contents of the addressed input/output channel on the GSE data busses.

INOTLD                      In/Out Load from GSE                      1 MCT

Accept a channel address from the Ground Support Equipment and write the contents of the GSE data busses into the addressed input/output channel.

FETCH                      Fetch from Memory to GSE                      2 MCT

Accept from the Ground Support Equipment a setting for either FB or EB and an address for the corresponding memory, and place the contents of the addresses location on the GSE data busses. Do not edit if the address is 0020-0023. Then restore b(BB).

STORE                      Store in Memory from GSE                      2 MCT

Accept from the Ground Support Equipment a setting for EB and an address in erasable memory, and write the contents of the GSE data busses into the addressed location. Then restore b(BB), unless the location stored into is BB itself.

The manual override instructions can occur at any time because they are not obliged to preserve the state of the computer.

GOJ                      Go    Jam                      2 MCT

Set  $c(Q) = b(Z)$ ;

Take next instruction from location 4000 and proceed from there.

TCSAJ                      Transfer Control to Specified Address Jam                      2 MCT

Take next instruction from the location whose address is on the Ground Support Equipment data busses, and proceed from there.

#### Address Constant Formats

The address constants available for Block II programming are considerably different than for Block I. A summary of them follows. The EBANK= code is also discussed.

|        |                 |
|--------|-----------------|
| ADRES  | Address         |
| REMADR | Remote Address  |
| GENADR | General Address |

Each of these codes creates a single precision constant word identical to the instruction word that would have resulted if the opcode had been TC. ADRES requires the location and address values to be in the same F - Bank if both are in F - Banks and to be in the same E - Bank if both are in E - Banks. REMADR requires the location and address values to be in different F - Banks if both are in F - Banks and to be in different E - Banks if both are in E - Banks. GENADR doesn't care.

CADR      FCADR                      (Fixed) Complete Address

These codes are synonymous. The address value must be in an F - Bank. The resulting single precision constant word equals the pseudo-address value minus octal 10000. Bits 15-11 equal the F - Bank number and bits 10 - 1 equal the relative location of the address in that bank.

ECADR                      Erasable Complete Address

The address value must be erasable, 0000-3777, and the resulting single precision word equals the the eleven bit pseudo-address. Bits 15-12 = 0.

EBANK=                      Erasable Bank Declaration

This code does not generate an AGC word. It informs the assembler of which E-Bank the programmer intends subsequent E-Bank addresses to be in. For basic instructions and interpretive address words,\*\* the assembler complains wherever an address is equivalent to a location in a different E-Bank. If the EBANK= code is followed by\* a BBCON, 2BCADR or 2CADR code, this EBANK= value is good only for that one subsequent code, and then the previous EBANK= setting is restored. This is called a "one-shot EBANK= declaration", and is required before every BBCON and 2BCADR or 2CADR.

---

\* "followed by" means with no instructions, interpretive opcode words, or address constants intervening.

\*\* Interpretive address EBANK restriction is for BLK2 programs but not for AGC programs.

SBANK=

Super Bank Declaration

Similar to EBANK=, but not required before BBCONs and 2CADRs. The address must be in a superbank.

1DNADR

1-Word Downlist Address

Same as ECADR, but used when the word addressed is the left half of a double-precision word for down telemetry.

2DNADR - 6DNADR

N-Word Downlist Address,  $2 \leq N \leq 6$

Same as 1DNADR, but with the 4 unused bits of the ECADR format filled in with 0001-0101. Used to point to a list of N double-precision words, stored consecutively, for down telemetry.

DNCHAN

Downlist Channel Address

Same as 1DNADR, but with prefix bits 0111. Used to point to a pair of channels for down telemetry.

DNPTR

Down Telemetry Sublist Pointer

Same as CAF, but tagged as a constant. Used in a downlist to point to a sublist.

BBCON

Both Bank Constant

This code generates a single precision constant word intended as data to be placed in the BB central register. The address value must be a fixed memory location or it must be equivalent to a valid F-Bank number, (range 0-27 now, 0-43 later). Bits 15-11 of the resulting word equal the address' bank number (fixed - fixed being banks 2 and 3). Bits 10 - 4 are zeros. Bits 3 - 1 equal the current EBANK= code.

2CADR      2BCADR

Double Complete Address Including  
a BBCON

These codes are synonymous. This code is intended to be used as the operand of a DTCB (DXCH Z) instruction. Two constant words are generated by this code. The first word is formed under the rules for GENADR. If the address value is in fixed memory, the second word is formed under the rules for BBCON. For an erasable address the second word becomes 0000x where x = the address' octal code EBANK number in the range 0 - 7.

2FCADR

Double Complete Address Including an  
FCADR

This code's address value must be in fixed memory. The code is intended as an operand of a DTCF (DXCH FB) instruction. Two constant words are generated by this code. The first word is formed under the rules for FCADR, and second under the rules for GENADR. Exception: both words are GENADRs if address value is in fixed fixed.

#### Control Pulse Definitions

To understand the control pulses and the pulse sequences, it is necessary to know the unaddressable central registers:

G                      Memory Local Register                      Bits 1 - 16

In an MCT in which erasable memory is cycled, the word from memory appears in G by the 5th microsecond (time 5 of 12 times) of the MCT. If it is left there through time 12, it is

restored exactly as it was read out. If a new value is written into G before time 10, that becomes the new value in the memory location. When fixed memory is cycled, the word appears in G by time 7.

WL                      Write Lines or Busses                      Bits 1 - 16  
These are the normal medium of communication among central registers, although some private lines exist.

B                      General Buffer Register                      Bits 1 - 16  
The B register always holds the instruction word at the beginning of each instruction.

C                      Complement Output of B                      Bits 1 - 16  
Not a separate storage. Each bit of C is the opposite of the corresponding bit of B.

Y                      Primary Adder Input                      Bits 1 - 16  
Has conventional and doubling inputs.

X                      Secondary Adder Input                      Bits 1 - 16  
Fed by private line from A and from constant generators.

U                      Adder Output                      Bits 1 - 16  
Exists as a function of  $c(X)$  and  $c(Y)$  - has no storage of its own.

S                      Address Selection Register                      Bits 1 - 12  
Holds the address of a fixed memory location from time 8 of the preceding MCT through time 7 of the current MCT, or holds (in bits 1 - 10) the address of an erasable memory location from time 1 through time 7 of an MCT.

SQ

Sequence Selection Register

Bits 10-16

Holds the operation code during execution of each instruction. Bit 15 is the extracode bit. SQ is aided by a three-bit stage counter and two branch flip-flops. A stage counter value of 2 selects the standard fetch-next-instruction subinstruction, regardless of the c(SQ) and the branch bits. Sequence selection by SQ is suppressed during counter interrupts by a signal called INKL.

# CONTROL PULSF DEFINITIONS

|        |   |                                                                                                                     |
|--------|---|---------------------------------------------------------------------------------------------------------------------|
| A2X    |   | COPY A1-16 INTO X1-16 BY PRIVATE LINE.                                                                              |
| R15X   |   | SET BIT 15 OF X TO 1.                                                                                               |
| CI     |   | INSERT CARRY INTO BIT 1 OF THE ADDER.                                                                               |
| CLXC   |   | CLEAR X CONDITIONAL ON THE OUTCOME OF TSGU. X IS<br>CLEARED IF BR1 = 0. USED IN DIVIDE.                             |
| DVST   |   | CAUSE DIVIDE STAGING BY A SIMPLE RULE. ALSO PERMIT<br>STAGING TO OCCUR AT TIMES3 OF DIVIDE CYCLES.                  |
| EXT    |   | SET THE EXTEND FLIP FLOP.                                                                                           |
| G2LS   | * | COPY G4-15,16,1 INTO L1-12,16,15.                                                                                   |
| KRPT   |   | RESET INTERRUPT PRIORITY CELL.                                                                                      |
| L16    |   | SET BIT 16 OF L TO 1.                                                                                               |
| L2GD   |   | COPY L1-14,16 INTO G2-15,16 -- ALSO MCRO INTO G1.                                                                   |
| MONEX  |   | SET BITS 2-16 OF X TO ONES.                                                                                         |
| MOUT   |   | NEGATIVE RATE OUTPUT PULSE.                                                                                         |
| NEACOF |   | PERMIT END AROUND CARRY AFTER END OF MP3.                                                                           |
| NEACON |   | INHIBIT END AROUND CARRY UNTIL NEACOF.                                                                              |
| NISQ   |   | NEXT INSTRUCTION IS TO BE LOADED INTO SQ. ALSO<br>FREES CERTAIN RESTRICTIONS- PERMITS INCRFMENTS AND<br>INTERRUPTS. |
| PIFL   |   | WHEN L15 = 1, BLOCK WRITING INTO Y1 ON A WYD.                                                                       |
| PONEX  |   | SET BIT 1 OF X TO 1.                                                                                                |
| POUT   |   | POSITIVE RATE OUTPUT PULSE.                                                                                         |
| PTWOX  |   | SET BIT 2 OF X TO 1.                                                                                                |
| R15    |   | PLACE OCTAL 000015 ON WL'S.                                                                                         |



# CONTROL PULSE DEFINITIONS

|        |                                                                                                                                                                                                                      |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R1C    | PLACE OCTAL 177776 = -1 ON WL'S.                                                                                                                                                                                     |
| R6     | PLACE OCTAL 000006 ON WRITE LINES.                                                                                                                                                                                   |
| RA     | READ A1-16 TO WL1-16.                                                                                                                                                                                                |
| RAD    | READ ADDRESS OF NEXT CYCLE. THIS APPEARS AT THE END OF AN INSTRUCTION AND NORMALLY IS INTERPRETED AS RG. IF THE NEXT INSTRUCTION IS TO BE A PSFUDO CODE (INHINT,RELINT,EXTEND), IT IS INSTEAD INTERPRETED AS RZ ST2. |
| RB     | READ B1-16 TO WL1-16.                                                                                                                                                                                                |
| RB1    | PLACE OCTAL 000001 ON THE WL'S.                                                                                                                                                                                      |
| RB1F   | PLACE OCTAL 000001 ON THE WL'S CONDITIONAL ON THE OUTCOME OF TSGU. RB1F IF BR1=1.                                                                                                                                    |
| RB2    | PLACE OCTAL 000002 ON THE WL'S.                                                                                                                                                                                      |
| RBBK   | READ THE BB (BOTH BANK) CONFIGURATION ONTO THE WRITE LINES. I.E. FB 9-11 TO WL 1-3 AND FB 11-14,16,16 TO WL 11-14,15,16.                                                                                             |
| RC     | READ THE CONTENT OF B INVERTED: C1-16 TO WL1-16.                                                                                                                                                                     |
| RCH    | READ THE CONTENT OF THE INPUT OR OUTPUT CHANNEL SPECIFIED BY THE CURRENT CONTENT OF S: CHANNEL BITS 1-14 TO WL1-14, AND BIT 16 TO WL15,16. CHANNELS 1 AND 2 READ AS RL AND RQ.                                       |
| RG     | READ G1-16 TO WL1-16.                                                                                                                                                                                                |
| RL     | READ L1-14 TO WL1-14, AND L16 TO WL15 AND 16.                                                                                                                                                                        |
| RL10RB | READ LOW 10 BITS OF B TO WL 1-10.                                                                                                                                                                                    |
| RQ     | READ Q1-16 TO WL1-16.                                                                                                                                                                                                |
| RRPA   | READ THE ADDRESS OF THE HIGHEST PRIORITY INTERRUPT REQUESTED.                                                                                                                                                        |
| RSC    | READ THE CONTENT OF CENTRAL STORE DEFINED BY THE ADDRESS CURRENTLY IN S: CENTRAL STORE BITS 1-16 ARE COPIED TO WL1-16.                                                                                               |
| RSCT   | READ THE ADDRESS OF HIGHEST PRIORITY COUNTER REQUEST                                                                                                                                                                 |
| RSTRT  | PLACE OCTAL 004000 = BLOCK 2 START ADDRESS ON WL'S.                                                                                                                                                                  |
| RSTSTG | RESET THE DIVIDE T03 STAGING CONDITION.                                                                                                                                                                              |
| RU     | READ U1-16 TO WL1-16.                                                                                                                                                                                                |

# CONTROL PULSF DEFINITIONS

|       |   |                                                                                                                                                                            |
|-------|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RUS   |   | READ U1-14 TO WL1-14, AND U15 TO WL15 AND 16.                                                                                                                              |
| RZ    |   | READ Z1-16 TO WL1-16.                                                                                                                                                      |
| ST1   |   | SET STAGF1 FLIP FLOP NEXT T12.                                                                                                                                             |
| ST2   |   | SET STAGF2 FLIP FLOP NEXT T12.                                                                                                                                             |
| STAGF |   | EXECUTE GRFY-CODED STAGE ADVANCE COMPUTED BY DVST.                                                                                                                         |
| TL15  |   | COPY L15 INTO BR1.                                                                                                                                                         |
| TMZ   |   | TEST WL1-16 FOR ALL ONES (-0). SET BR2 IF TRUE.                                                                                                                            |
| TOV   |   | TEST FOR + OR - OVERFLOW. SET BR1,2 TO 00 IF NO OVERFLOW, 01 IF + OVERFLOW, 10 IF - OVERFLOW.                                                                              |
| TPZG  |   | TEST CONTENT OF G FOR PLUS ZERO. IF TRUE SET BR2=1.                                                                                                                        |
| TRSM  |   | TEST FOR RESUME ADDRESS ON INDEX. ST2 IF (S)=0017.                                                                                                                         |
| TSGN  |   | TEST SIGN. COPY WL16 TO BR1.                                                                                                                                               |
| TSGN2 |   | TEST SIGN. COPY WL16 TO BR2.                                                                                                                                               |
| TSGU  |   | TEST SIGN OF SUM (U). COPY U16 INTO BR1.                                                                                                                                   |
| U2BBK |   | ADDER BITS 1-3 AND 11-14,16 ARE TRANSFERRED INTO ERASABLE AND FIXED BANKS. THIS PULSF MAY BE INHIBITED BY CTS SIGNAL MONWBK.                                               |
| WA    |   | CLEAR AND WRITE WL1-16 INTO A1-16.                                                                                                                                         |
| WALS  | * | CLEAR AND WRITE INTO A1-14 FROM WL3-16. CLEAR AND WRITE INTO L13,14 FROM WL1,2. CLEAR AND WRITE INTO A15,16 FROM G16 (IF G1=0) OR FROM WL16 (IF G1=1).                     |
| WB    |   | CLEAR AND WRITE WL1-16 INTO R1-16.                                                                                                                                         |
| WCH   |   | CLEAR AND WRITE WL1-14,16,PARITY INTO CHANNEL BITS 1-14,16,PARITY. CHANNELS 1 AND 2 WRITE AS WL AND WQ. THE CHANNEL TO BE LOADED IS SPECIFIED BY THE CURRENT CONTENT OF S. |
| WG    |   | CLEAR AND WRITE WL1-16 INTO G1-16 EXCEPT FOR ADDRESSES OCTAL 20-23, WHICH CAUSE EDITING.                                                                                   |
| WL    |   | CLEAR AND WRITE WL1-16 INTO L1-16.                                                                                                                                         |

# CONTROL PULSE DEFINITIONS

|      |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |      |       |       |          |       |       |          |   |   |   |   |    |   |   |   |   |   |    |    |   |   |   |   |   |    |     |   |   |   |   |   |    |    |    |      |   |   |   |    |    |   |   |   |   |   |    |     |   |   |   |   |   |    |    |    |      |   |   |   |   |    |   |      |
|------|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|-------|----------|-------|-------|----------|---|---|---|---|----|---|---|---|---|---|----|----|---|---|---|---|---|----|-----|---|---|---|---|---|----|----|----|------|---|---|---|----|----|---|---|---|---|---|----|-----|---|---|---|---|---|----|----|----|------|---|---|---|---|----|---|------|
| WOVR |    | TEST FOR OVERFLOW DURING COUNTER INCREMENTS AND PROGRAM INITIATED INCREMENTS (INCR AND AUG). RPT IF OVERFLOW OCCURS WHEN ADDRESSING CERTAIN COUNTERS.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |      |       |       |          |       |       |          |   |   |   |   |    |   |   |   |   |   |    |    |   |   |   |   |   |    |     |   |   |   |   |   |    |    |    |      |   |   |   |    |    |   |   |   |   |   |    |     |   |   |   |   |   |    |    |    |      |   |   |   |   |    |   |      |
| WQ   |    | CLFAR AND WRITE WL1-16 INTO Q1-16.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |      |       |       |          |       |       |          |   |   |   |   |    |   |   |   |   |   |    |    |   |   |   |   |   |    |     |   |   |   |   |   |    |    |    |      |   |   |   |    |    |   |   |   |   |   |    |     |   |   |   |   |   |    |    |    |      |   |   |   |   |    |   |      |
| WS   |    | CLFAR AND WRITE WL1-12 INTO S1-12.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |      |       |       |          |       |       |          |   |   |   |   |    |   |   |   |   |   |    |    |   |   |   |   |   |    |     |   |   |   |   |   |    |    |    |      |   |   |   |    |    |   |   |   |   |   |    |     |   |   |   |   |   |    |    |    |      |   |   |   |   |    |   |      |
| WSC  |    | CLFAR AND WRITE WL1-16 INTO THE CENTRAL REGISTER SPECIFIED BY THE CURRENT CONTENT OF S. BITS 1-16 INTO POSITIONS 1-16.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |      |       |       |          |       |       |          |   |   |   |   |    |   |   |   |   |   |    |    |   |   |   |   |   |    |     |   |   |   |   |   |    |    |    |      |   |   |   |    |    |   |   |   |   |   |    |     |   |   |   |   |   |    |    |    |      |   |   |   |   |    |   |      |
| WSQ  | *  | CLEAR AND WRITE WL10-14,16 INTO SQ10-14,16. AND COP THE EXTEND FLIP FLOP INTO SQ15.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |      |       |       |          |       |       |          |   |   |   |   |    |   |   |   |   |   |    |    |   |   |   |   |   |    |     |   |   |   |   |   |    |    |    |      |   |   |   |    |    |   |   |   |   |   |    |     |   |   |   |   |   |    |    |    |      |   |   |   |   |    |   |      |
| WY   |    | CLEAR Y AND X. WRITE WL1-16 INTO Y1-16.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |      |       |       |          |       |       |          |   |   |   |   |    |   |   |   |   |   |    |    |   |   |   |   |   |    |     |   |   |   |   |   |    |    |    |      |   |   |   |    |    |   |   |   |   |   |    |     |   |   |   |   |   |    |    |    |      |   |   |   |   |    |   |      |
| WY12 |    | CLFAR Y AND X. WRITE WL1-12 INTO Y1-12.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |      |       |       |          |       |       |          |   |   |   |   |    |   |   |   |   |   |    |    |   |   |   |   |   |    |     |   |   |   |   |   |    |    |    |      |   |   |   |    |    |   |   |   |   |   |    |     |   |   |   |   |   |    |    |    |      |   |   |   |   |    |   |      |
| WYD  |    | CLEAR Y AND X. WRITE WL1-14 INTO Y2-15. WRITE WL16 INTO Y16. WRITE WL16 INTO Y1 EXCEPT:<br>(1) WHEN END-AROUND CARRY IS INHIBITED BY NEACON,<br>(2) DURING SHINC SEQUENCE, OR<br>(3) PIFL IS ACTIVE AND L15 = 1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |      |       |       |          |       |       |          |   |   |   |   |    |   |   |   |   |   |    |    |   |   |   |   |   |    |     |   |   |   |   |   |    |    |    |      |   |   |   |    |    |   |   |   |   |   |    |     |   |   |   |   |   |    |    |    |      |   |   |   |   |    |   |      |
| WZ   |    | CLFAR AND WRITE WL1-16 INTO Z1-16.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |      |       |       |          |       |       |          |   |   |   |   |    |   |   |   |   |   |    |    |   |   |   |   |   |    |     |   |   |   |   |   |    |    |    |      |   |   |   |    |    |   |   |   |   |   |    |     |   |   |   |   |   |    |    |    |      |   |   |   |   |    |   |      |
| Z15  |    | SET BIT 15 OF Z TO 1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |      |       |       |          |       |       |          |   |   |   |   |    |   |   |   |   |   |    |    |   |   |   |   |   |    |     |   |   |   |   |   |    |    |    |      |   |   |   |    |    |   |   |   |   |   |    |     |   |   |   |   |   |    |    |    |      |   |   |   |   |    |   |      |
| Z16  |    | SET BIT 16 OF Z TO 1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |      |       |       |          |       |       |          |   |   |   |   |    |   |   |   |   |   |    |    |   |   |   |   |   |    |     |   |   |   |   |   |    |    |    |      |   |   |   |    |    |   |   |   |   |   |    |     |   |   |   |   |   |    |    |    |      |   |   |   |   |    |   |      |
| ZAP  |    | ALWAYS IMPLIFS RU, G2LS, AND WAL5.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |      |       |       |          |       |       |          |   |   |   |   |    |   |   |   |   |   |    |    |   |   |   |   |   |    |     |   |   |   |   |   |    |    |    |      |   |   |   |    |    |   |   |   |   |   |    |     |   |   |   |   |   |    |    |    |      |   |   |   |   |    |   |      |
| ZIP  |    | ALWAYS IMPLIFS A2X AND L2GD. ALSO IF L15,2,1 ARE:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |      |       |       |          |       |       |          |   |   |   |   |    |   |   |   |   |   |    |    |   |   |   |   |   |    |     |   |   |   |   |   |    |    |    |      |   |   |   |    |    |   |   |   |   |   |    |     |   |   |   |   |   |    |    |    |      |   |   |   |   |    |   |      |
|      |    | <table><tr><td>L15</td><td>L2</td><td>L1</td><td>READ</td><td>WRITE</td><td>CARRY</td><td>REMEMBER</td></tr><tr><td>0</td><td>0</td><td>0</td><td>-</td><td>WY</td><td>-</td><td>-</td></tr><tr><td>0</td><td>0</td><td>1</td><td>RB</td><td>WY</td><td>-</td><td>-</td></tr><tr><td>0</td><td>1</td><td>0</td><td>RB</td><td>WYD</td><td>-</td><td>-</td></tr><tr><td>0</td><td>1</td><td>1</td><td>RC</td><td>WY</td><td>CI</td><td>MCRO</td></tr><tr><td>1</td><td>0</td><td>0</td><td>RB</td><td>WY</td><td>-</td><td>-</td></tr><tr><td>1</td><td>0</td><td>1</td><td>RB</td><td>WYD</td><td>-</td><td>-</td></tr><tr><td>1</td><td>1</td><td>0</td><td>RC</td><td>WY</td><td>CI</td><td>MCRO</td></tr><tr><td>1</td><td>1</td><td>1</td><td>-</td><td>WY</td><td>-</td><td>MCRO</td></tr></table> | L15  | L2    | L1    | READ     | WRITE | CARRY | REMEMBER | 0 | 0 | 0 | - | WY | - | - | 0 | 0 | 1 | RB | WY | - | - | 0 | 1 | 0 | RB | WYD | - | - | 0 | 1 | 1 | RC | WY | CI | MCRO | 1 | 0 | 0 | RB | WY | - | - | 1 | 0 | 1 | RB | WYD | - | - | 1 | 1 | 0 | RC | WY | CI | MCRO | 1 | 1 | 1 | - | WY | - | MCRO |
| L15  | L2 | L1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | READ | WRITE | CARRY | REMEMBER |       |       |          |   |   |   |   |    |   |   |   |   |   |    |    |   |   |   |   |   |    |     |   |   |   |   |   |    |    |    |      |   |   |   |    |    |   |   |   |   |   |    |     |   |   |   |   |   |    |    |    |      |   |   |   |   |    |   |      |
| 0    | 0  | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | -    | WY    | -     | -        |       |       |          |   |   |   |   |    |   |   |   |   |   |    |    |   |   |   |   |   |    |     |   |   |   |   |   |    |    |    |      |   |   |   |    |    |   |   |   |   |   |    |     |   |   |   |   |   |    |    |    |      |   |   |   |   |    |   |      |
| 0    | 0  | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | RB   | WY    | -     | -        |       |       |          |   |   |   |   |    |   |   |   |   |   |    |    |   |   |   |   |   |    |     |   |   |   |   |   |    |    |    |      |   |   |   |    |    |   |   |   |   |   |    |     |   |   |   |   |   |    |    |    |      |   |   |   |   |    |   |      |
| 0    | 1  | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | RB   | WYD   | -     | -        |       |       |          |   |   |   |   |    |   |   |   |   |   |    |    |   |   |   |   |   |    |     |   |   |   |   |   |    |    |    |      |   |   |   |    |    |   |   |   |   |   |    |     |   |   |   |   |   |    |    |    |      |   |   |   |   |    |   |      |
| 0    | 1  | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | RC   | WY    | CI    | MCRO     |       |       |          |   |   |   |   |    |   |   |   |   |   |    |    |   |   |   |   |   |    |     |   |   |   |   |   |    |    |    |      |   |   |   |    |    |   |   |   |   |   |    |     |   |   |   |   |   |    |    |    |      |   |   |   |   |    |   |      |
| 1    | 0  | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | RB   | WY    | -     | -        |       |       |          |   |   |   |   |    |   |   |   |   |   |    |    |   |   |   |   |   |    |     |   |   |   |   |   |    |    |    |      |   |   |   |    |    |   |   |   |   |   |    |     |   |   |   |   |   |    |    |    |      |   |   |   |   |    |   |      |
| 1    | 0  | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | RB   | WYD   | -     | -        |       |       |          |   |   |   |   |    |   |   |   |   |   |    |    |   |   |   |   |   |    |     |   |   |   |   |   |    |    |    |      |   |   |   |    |    |   |   |   |   |   |    |     |   |   |   |   |   |    |    |    |      |   |   |   |   |    |   |      |
| 1    | 1  | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | RC   | WY    | CI    | MCRO     |       |       |          |   |   |   |   |    |   |   |   |   |   |    |    |   |   |   |   |   |    |     |   |   |   |   |   |    |    |    |      |   |   |   |    |    |   |   |   |   |   |    |     |   |   |   |   |   |    |    |    |      |   |   |   |   |    |   |      |
| 1    | 1  | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | -    | WY    | -     | MCRO     |       |       |          |   |   |   |   |    |   |   |   |   |   |    |    |   |   |   |   |   |    |     |   |   |   |   |   |    |    |    |      |   |   |   |    |    |   |   |   |   |   |    |     |   |   |   |   |   |    |    |    |      |   |   |   |   |    |   |      |
| ZOUT |    | NO RATE OUTPUT PULSE. RESET OUTBIT REQUESTING DINC.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |      |       |       |          |       |       |          |   |   |   |   |    |   |   |   |   |   |    |    |   |   |   |   |   |    |     |   |   |   |   |   |    |    |    |      |   |   |   |    |    |   |   |   |   |   |    |     |   |   |   |   |   |    |    |    |      |   |   |   |   |    |   |      |

\* THESE PULSES DO NOT APPEAR IN THE PULSE SEQUENCES.

# PROGRAMMABLE INSTRUCTIONS

| OP CODE     | EXT | 5016.14-10 | OPERATION                                |
|-------------|-----|------------|------------------------------------------|
| TC          | 0   | 000        | TRANSFER CONTROL AND PSEUDO-CODES *      |
| CCS         | 0   | 001 00     | COUNT, COMPARE, AND SKIP                 |
| TCF         | 0   | 001 01     | TRANSFER CONTROL TO FIXED                |
| TCF         | 0   | 001 10     | " " " "                                  |
| TCF         | 0   | 001 11     | " " " "                                  |
| DAS         | 0   | 010 00     | DOUBLE PRECISION ADD TO STORAGE          |
| LXCH        | 0   | 010 01     | L EXCHANGE WITH MEMORY                   |
| INCR        | 0   | 010 10     | INCREMENT MEMORY                         |
| ADS         | 0   | 010 11     | ADD TO STORAGE                           |
| CA          | 0   | 011        | CLEAR AND ADD                            |
| CS          | 0   | 100        | CLEAR AND SUBTRACT                       |
| INDEX (NDX) | 0   | 101 00     | INDEX NEXT INSTRUCTION (INDEX 17=RESUME) |
| DXCH        | 0   | 101 01     | DOUBLE PRECISION EXCHANGE WITH MEMORY    |
| TS          | 0   | 101 10     | TRANSFER TO STORAGE                      |
| XCH         | 0   | 101 11     | EXCHANGE WITH MEMORY                     |
| AD          | 0   | 110        | ADD                                      |
| MASK (MSK)  | 0   | 111        | MASK ("AND" TO A)                        |
| READ        | 1   | 000 00 0   | READ FROM CHANNEL                        |
| WRITE       | 1   | 000 00 1   | WRITE IN CHANNEL                         |
| RAND        | 1   | 000 01 0   | READ, "AND" TO A                         |
| WAND        | 1   | 000 01 1   | WRITE, "AND" TO CHANNEL                  |
| ROR         | 1   | 000 10 0   | READ, "OR" TO A                          |
| WOR         | 1   | 000 10 1   | WRITE, "OR" TO CHANNEL                   |
| RXOR        | 1   | 000 11 0   | READ, EXCLUSIVE "OR" TO A                |
| EDRUPT      | 1   | 000 11 1   | ED SMALLY'S OWN RUPT ORDER               |
| DV          | 1   | 001 00     | DIVIDE                                   |
| BZF         | 1   | 001 01     | BRANCH ON ZERO TO FIXED                  |
| BZF         | 1   | 001 10     | " " " " "                                |
| BZF         | 1   | 001 11     | " " " " "                                |
| MSU         | 1   | 010 00     | MODULAR SUBTRACT                         |
| QXCH        | 1   | 010 01     | Q EXCHANGE WITH MEMORY                   |
| AUG         | 1   | 010 10     | AUGMENT MEMORY                           |
| DIM         | 1   | 010 11     | DIMINISH MEMORY                          |
| DCA         | 1   | 011        | DOUBLE PRECISION CLEAR AND ADD           |
| DCS         | 1   | 100        | DOUBLE PRECISION CLEAR AND SUBTRACT      |
| INDEX (NDX) | 1   | 101        | INDEX NEXT EXTRACODE INSTRUCTION         |
| SU          | 1   | 110 00     | SUBTRACT                                 |
| BZMF        | 1   | 110 01     | BRANCH ON ZERO OR MINUS TO FIXED         |
| BZMF        | 1   | 110 10     | " " " " " " "                            |
| BZMF        | 1   | 110 11     | " " " " " " "                            |
| MP          | 1   | 111        | MULTIPLY                                 |

\* PSEUDO-CODES: RELINT = TC 0003, INHINT = TC 0004, EXTEND = TC 0006. THE TC OPERATION CODE IS SHARED BY THE NON-PROGRAMMABLE SEQUENCES GOJ1 (FOLLOWED BY TCO) AND TCSAJ3 (FOLLOWED BY STD2).

# PULSE SEQUENCES

## TC0

|    |             |
|----|-------------|
| 1. | RP WY12 CI  |
| 2. | RSC WG NISQ |
| 3. | RZ WQ       |
| 6. | RU WZ       |
| 8. | RAD WB WS   |

## GOJ1

|    |             |
|----|-------------|
| 2. | RSC WG      |
| 8. | RSTRT WS WB |

## TCSAJ3

|    |           |
|----|-----------|
| 2. | RSC WG    |
| 8. | WS WZ ST2 |

## CCS0

|        |                     |
|--------|---------------------|
| 1.     | RI 10BR WS          |
| 2.     | RSC WG              |
| 5.     | RG WB TSGN TMZ TP7G |
| 7. 00  | RZ WY12             |
| 7. 01  | RZ WY12 PONFX       |
| 7. 10  | RZ WY12 PTWOX       |
| 7. 11  | RZ WY12 PONFX PTWOX |
| 8.     | RU WZ WS            |
| 9.     | RB WG               |
| 10. 00 | RR WY MONFX CI ST2  |
| 10. X1 | WY ST2              |
| 10. 10 | RC WY MONFX CI ST2  |
| 11.    | RU WA               |

## TCF0

|    |             |
|----|-------------|
| 1. | RB WY12 CI  |
| 2. | RSC WG NISQ |
| 6. | RU WZ       |
| 8. | RAD WB WS   |

# PULSE SEQUENCES

## DAS0

|        |                         |
|--------|-------------------------|
| 1.     | RL10BB WS WY12 MONEX CI |
| 2.     | RSC WG                  |
| 3.     | RA WB                   |
| 4.     | RI WA                   |
| 5.     | RU WL                   |
| 6.     | RG WY A2X               |
| 7.     | RR WA                   |
| 8.     | RL WB                   |
| 9.     | RU WSC WG TOV           |
| 10. 00 | RA WY ST1               |
| 10. 01 | RA WY ST1 PONEX         |
| 10. 10 | RA WY ST1 MONEX         |
| 10. 11 | RA WY ST1               |

## DAS1

|        |               |
|--------|---------------|
| 1.     | RL10BB WS     |
| 2.     | RSC WG        |
| 3.     | RU WA         |
| 5.     | RG WY A2X     |
| 6.     | RU WG WSC TOV |
| 7. 00  | WA            |
| 7. 01  | WA RB1        |
| 7. 10  | WA R1C        |
| 7. 11  | WA            |
| 8.     | RZ WS ST2     |
| 9.     | RC TMZ        |
| 10. X0 | WL            |
| 11. X1 | RU WA         |

## LXCH0

|    |           |
|----|-----------|
| 1. | RL10BB WS |
| 2. | RSC WG    |
| 3. | RL WB     |
| 5. | RG WL     |
| 7. | RR WSC WG |
| 8. | RZ WS ST2 |

## INCRO

|    |                     |
|----|---------------------|
| 1. | RL10BB WS           |
| 2. | RSC WG              |
| 5. | RG WY TSGN TMZ TPZG |
| 6. | PONEX               |
| 7. | RU WSC WG WOVR      |
| 8. | RZ WS ST2           |

## PULSF SEQUENCES

### ADSO

|     |    |               |
|-----|----|---------------|
| 1.  |    | RL10BB WS     |
| 2.  |    | RSC WG        |
| 5.  |    | RG WY A2X     |
| 6.  |    | RU WSC WG TOV |
| 7.  | 00 | WA            |
| 7.  | 01 | WA PB1        |
| 7.  | 10 | WA R1C        |
| 7.  | 11 | WA            |
| 8.  |    | RZ WS ST2     |
| 9.  |    | RC TMZ        |
| 11. |    | RU WA         |

### CA0

|     |  |           |
|-----|--|-----------|
| 2.  |  | RSC WG    |
| 7.  |  | RG WB     |
| 8.  |  | RZ WS ST2 |
| 9.  |  | RB WG     |
| 10. |  | RB WA     |

### CS0

|     |  |           |
|-----|--|-----------|
| 2.  |  | RSC WG    |
| 7.  |  | RG WB     |
| 8.  |  | RZ WS ST2 |
| 9.  |  | RB WG     |
| 10. |  | RC WA     |

## PULSF SEQUENCES

### NDX0

|     |        |
|-----|--------|
| 2.  | RSC WG |
| 5.  | TRSM   |
| 7.  | RG WB  |
| 8.  | RZ WS  |
| 9.  | RB WG  |
| 10. | ST1    |

### NDX1

|     |             |
|-----|-------------|
| 1.  | RZ WY12 CI  |
| 2.  | RSC WG NISQ |
| 3.  | RP WZ       |
| 4.  | RA WB       |
| 5.  | RZ WA       |
| 6.  | RU WZ       |
| 7.  | RG WY A2X   |
| 8.  | RU WS       |
| 9.  | RB WA       |
| 10. | RU WB       |

### RSM3

|    |             |
|----|-------------|
| 1. | R15 WS      |
| 2. | RSC WG NISQ |
| 5. | RG WZ       |
| 6. | RB WG       |
| 8. | RAD WP WS   |

### DXCH0

|     |                         |
|-----|-------------------------|
| 1.  | RL10BB WS WY12 MOMEX CI |
| 2.  | RSC WG                  |
| 3.  | RL WB                   |
| 5.  | RG WL                   |
| 7.  | RB WSC WG               |
| 8.  | RU WS WB                |
| 10. | ST1                     |

### DXCH1

|    |           |
|----|-----------|
| 1. | RL10BB WS |
| 2. | RSC WG    |
| 3. | RA WB     |
| 5. | RG WA     |
| 7. | RB WSC WG |
| 8. | RZ WS ST2 |



# PULSF SEQUENCES

## TSO

|    |    |            |
|----|----|------------|
| 1. |    | RI 10BR WS |
| 2. |    | RSC WG     |
| 3. |    | RA WB TOV  |
| 4. | 00 | RZ WY12    |
| 4. | 01 | RZ WY12 CI |
| 4. | 10 | RZ WY12 CI |
| 4. | 11 | RZ WY12    |
| 5. | 01 | RB1 WA     |
| 5. | 10 | R1C WA     |
| 6. |    | RU WZ      |
| 7. |    | RB WSC WG  |
| 8. |    | RZ WS ST2  |

## XCHO

|    |  |           |
|----|--|-----------|
| 1. |  | RL10BR WS |
| 2. |  | RSC WG    |
| 3. |  | RA WB     |
| 5. |  | RG WA     |
| 7. |  | RB WSC WG |
| 8. |  | RZ WS ST2 |

## ADO

|     |  |           |
|-----|--|-----------|
| 2.  |  | RSC WG    |
| 7.  |  | RG WB     |
| 8.  |  | RZ WS ST2 |
| 9.  |  | RB WG     |
| 10. |  | RB WY A2X |
| 11. |  | RU WA     |

## MSKO

|     |  |           |
|-----|--|-----------|
| 2.  |  | RSC WG    |
| 3.  |  | RA WB     |
| 4.  |  | RC WA     |
| 7.  |  | RG WB     |
| 8.  |  | RZ WS ST2 |
| 9.  |  | RC RA WY  |
| 10. |  | RU WB     |
| 11. |  | RC WA     |

## PULSE SEQUENCES

### READO

|    |           |
|----|-----------|
| 1. | RL10BB WS |
| 2. | RA WB     |
| 3. | WY        |
| 4. | RCH WR    |
| 5. | RB WA     |
| 6. | RA WB     |
| 8. | RZ WS ST2 |

### WRITEO

|    |           |
|----|-----------|
| 1. | RL10BB WS |
| 2. | RA WB WG  |
| 3. | WY        |
| 4. | RCH WR    |
| 5. | RA WCH    |
| 6. | RA WB     |
| 8. | RZ WS ST2 |

### RANDO

|    |           |
|----|-----------|
| 1. | RL10BB WS |
| 2. | RA WB     |
| 3. | RC WY     |
| 4. | RCH WR    |
| 5. | RC RU WA  |
| 6. | RA WB     |
| 7. | RC WA     |
| 8. | RZ WS ST2 |

### WANDO

|    |           |
|----|-----------|
| 1. | RL10BB WS |
| 2. | RA WB     |
| 3. | RC WY     |
| 4. | RCH WR    |
| 5. | RC RU WA  |
| 6. | RA WB     |
| 7. | RC WA WCH |
| 8. | RZ WS ST2 |

## PULSF SEQUENCES

### RORO

|    |            |
|----|------------|
| 1. | RI 10BR WS |
| 2. | RA WB      |
| 3. | RB WY      |
| 4. | RCH WB     |
| 5. | RB RU WA   |
| 6. | RA WB      |
| 8. | RZ WS ST2  |

### WORD

|    |              |
|----|--------------|
| 1. | RL10BR WS    |
| 2. | RA WB        |
| 3. | RB WY        |
| 4. | RCH WB       |
| 5. | RB RU WA WCH |
| 6. | RA WB        |
| 8. | RZ WS ST2    |

### RXORO

|     |           |
|-----|-----------|
| 1.  | RL10BR WS |
| 2.  | RA WB     |
| 3.  | RC RCH WY |
| 4.  | RCH WB    |
| 5.  | RA RC WG  |
| 7.  | RG WB     |
| 8.  | RZ WS ST2 |
| 9.  | RC WG     |
| 10. | RI WB     |
| 11. | RC RG WA  |

### RUPT0

|     |        |
|-----|--------|
| 1.  | R15 WS |
| 2.  | RSC WG |
| 9.  | RZ WG  |
| 10. | ST1    |

### RUPT1

|    |            |
|----|------------|
| 1. | R15 RB2 WS |
| 2. | RSC WG     |
| 3. | RRPA WZ    |
| 8. | RZ WS ST2  |
| 9. | RB WG KRPT |

# PULSE SEQUENCES

## DV0

|    |    |                |
|----|----|----------------|
| 1. |    | RA WB TSGN TMZ |
| 2. | OX | RC WA TMZ DVST |
| 2. | 1X | DVST           |
| 3. |    | RU WB STAGE    |

## DV1

|     |    |                      |
|-----|----|----------------------|
| 4.  | X0 | RI WB                |
| 4.  | X1 | RI WB TSGN           |
| 5.  | OX | RR WY B15X           |
| 5.  | 1X | RC WY B15X Z16       |
| 6.  |    | RU WL TOV            |
| 7.  |    | RG RSC WB TSGN       |
| 8.  | X0 | RA WY PONEX          |
| 8.  | X1 | RA WY                |
| 9.  | OX | RR WA                |
| 9.  | 1X | RC WA Z15            |
| 10. |    | RU WB                |
| 11. |    | RL WYD               |
| 12. |    | RU WL                |
| 1.  |    | L2GD RB WYD A2X PIFL |
| 2.  | OX | RG WL TSGU DVST CLXC |
| 2.  | 1X | RG WL TSGU DVST RB1F |
| 3.  |    | RU WB STAGE          |

## DV3

|     |    |                      |
|-----|----|----------------------|
| 4.  |    | L2GD RB WYD A2X PIFL |
| 5.  | OX | RG WL TSGU CLXC      |
| 5.  | 1X | RG WL TSGU RB1F      |
| 6.  |    | RU WB                |
| 7.  |    | L2GD RB WYD A2X PIFL |
| 8.  | OX | RG WL TSGU CLXC      |
| 8.  | 1X | RG WL TSGU RB1F      |
| 9.  |    | RU WB                |
| 10. |    | L2GD RB WYD A2X PIFL |
| 11. | OX | RG WL TSGU CLXC      |
| 11. | 1X | RG WL TSGU RB1F      |
| 12. |    | RU WB                |
| 1.  |    | L2GD RB WYD A2X PIFL |
| 2.  | OX | RG WL TSGU DVST CLXC |
| 2.  | 1X | RG WL TSGU DVST RB1F |
| 3.  |    | RU WB STAGE          |

# PULSE SEQUENCES

## DV7

|     |    |                      |
|-----|----|----------------------|
| 4.  |    | L2GD RB WYD A2X PIFL |
| 5.  | 0X | RG WL TSGU CLXC      |
| 5.  | 1X | RG WL TSGU RB1F      |
| 6.  |    | RU WB                |
| 7.  |    | L2GD RB WYD A2X PIFL |
| 8.  | 0X | RG WL TSGU CLXC      |
| 8.  | 1X | RG WL TSGU RB1F      |
| 9.  |    | RU WB                |
| 10. |    | L2GD RB WYD A2X PIFL |
| 11. | 0X | RG WL TSGU CLXC      |
| 11. | 1X | RG WL TSGU RB1F      |
| 12. |    | RU WB                |
| 1.  |    | L2GD RB WYD A2X PIFL |
| 2.  | 0X | RG WL TSGU DVST CLXC |
| 2.  | 1X | RG WL TSGU DVST RB1F |
| 3.  |    | RU WB STAGE          |

## DV6

|     |    |                      |
|-----|----|----------------------|
| 4.  |    | L2GD RB WYD A2X PIFL |
| 5.  | 0X | RG WL TSGU CLXC      |
| 5.  | 1X | RG WL TSGU RB1F      |
| 6.  |    | RU WB                |
| 7.  |    | L2GD RB WYD A2X PIFL |
| 8.  | 0X | RG WL TSGU CLXC      |
| 8.  | 1X | RG WL TSGU RB1F      |
| 9.  |    | RU WB                |
| 10. |    | L2GD RB WYD A2X PIFL |
| 11. | 0X | RG WL TSGU CLXC      |
| 11. | 1X | RG WL TSGU RB1F      |
| 12. |    | RU WB                |
| 1.  |    | L2GD RB WYD A2X PIFL |
| 2.  | 0X | RG WL TSGU DVST CLXC |
| 2.  | 1X | RG WL TSGU DVST RB1F |
| 3.  |    | RU WB STAGE          |

## DV4

|     |    |                       |
|-----|----|-----------------------|
| 3.  |    | RU WB STAGE           |
| 4.  |    | L2GD RB WYD A2X PIFL  |
| 5.  | 0X | RG WB WA TSGU CLXC    |
| 5.  | 1X | RG WB WA TSGU RB1F    |
| 6.  |    | RZ TOV                |
| 7.  | 01 | RC WA                 |
| 7.  | 1X | RC WA                 |
| 8.  |    | RZ WS ST2 TSGN RSTSTG |
| 9.  |    | RU WB WL              |
| 10. | 0X | RC WL                 |

# PULSE SEQUENCES

## BZF0

|    |    |                |
|----|----|----------------|
| 1. |    | RA WG TSGN TMZ |
| 2. |    | TPZG           |
| 3. |    | RSC WG         |
| 5. | X1 | RR WY12 CI     |
| 6. | X1 | RU WZ          |
| 8. | X0 | RZ WS ST2      |
| 8. | X1 | RAD WB WS NISO |

## MSU0

|     |    |              |
|-----|----|--------------|
| 1.  |    | RL10BB WS    |
| 2.  |    | RSC WG       |
| 5.  |    | RG WB        |
| 6.  |    | RC WY CI A2X |
| 7.  |    | RUS WA TSGN  |
| 8.  |    | RZ WS ST2    |
| 9.  |    | RB WG        |
| 10. | 1X | RA WY MONEX  |
| 11. |    | RUS WA       |

## QXCH0

|    |  |           |
|----|--|-----------|
| 1. |  | RL10BB WS |
| 2. |  | RSC WG    |
| 3. |  | RO WB     |
| 5. |  | RG WQ     |
| 7. |  | RR WSC WG |
| 8. |  | RZ WS ST2 |

## AUG0

|    |    |                     |
|----|----|---------------------|
| 1. |    | RL10BB WS           |
| 2. |    | RSC WG              |
| 5. |    | RG WY TSGN TMZ TPZG |
| 6. | 0X | PONEX               |
| 6. | 1X | MONEX               |
| 7. |    | RU WSC WG WOVF      |
| 8. |    | RZ WS ST2           |

## DIMO

|    |    |                     |
|----|----|---------------------|
| 1. |    | RL10BB WS           |
| 2. |    | RSC WG              |
| 5. |    | RG WY TSGN TMZ TPZG |
| 6. | 00 | MONEX               |
| 6. | 10 | PONEX               |
| 7. |    | RU WSC WG WOVF      |
| 8. |    | RZ WS ST2           |

## PULSF SEQUENCES

### DCA0

|     |                  |
|-----|------------------|
| 1.  | RR WY12 MONEX CI |
| 2.  | RSC WG           |
| 7.  | RG WB            |
| 8.  | RU WS            |
| 9.  | RP WG            |
| 10. | RP WL ST1        |

### DCA1

|     |           |
|-----|-----------|
| 2.  | RSC WG    |
| 7.  | RG WB     |
| 8.  | RZ WS ST2 |
| 9.  | RP WG     |
| 10. | RB WA     |

### DCS0

|     |                  |
|-----|------------------|
| 1.  | RR WY12 MONEX CI |
| 2.  | RSC WG           |
| 7.  | RG WB            |
| 8.  | RU WS            |
| 9.  | RB WG            |
| 10. | RC WL ST1        |

### DCS1

|     |           |
|-----|-----------|
| 2.  | RSC WG    |
| 7.  | RG WB     |
| 8.  | RZ WS ST2 |
| 9.  | RP WG     |
| 10. | RC WA     |

# PULSE SEQUENCES

## NDXX0

|     |        |
|-----|--------|
| 2.  | RSC WG |
| 7.  | RG WB  |
| 8.  | R7 WS  |
| 9.  | RB WG  |
| 10. | ST1    |

## NDXX1

|     |             |
|-----|-------------|
| 1.  | RZ WY12 CI  |
| 2.  | RSC WG NISO |
| 3.  | RB WZ       |
| 4.  | RA WB       |
| 5.  | RZ WA       |
| 6.  | RU WZ       |
| 7.  | RG WY A2X   |
| 8.  | RU WS       |
| 9.  | RB WA       |
| 10. | RU WB EXT   |

## SU0

|     |           |
|-----|-----------|
| 2.  | RSC WG    |
| 7.  | RG WB     |
| 8.  | RZ WS ST2 |
| 9.  | RB WG     |
| 10. | RG WY A2X |
| 11. | RU WA     |

## BZMFO

|    |                   |
|----|-------------------|
| 1. | RA WG TSGN TMZ    |
| 2. | TPZG              |
| 3. | RSC WG            |
| 5. | 01 RB WY12 CI     |
| 5. | 10 RB WY12 CI     |
| 5. | 11 RB WY12 CI     |
| 6. | 01 RU WZ          |
| 6. | 10 RU WZ          |
| 6. | 11 RU WZ          |
| 8. | 00 RZ WS ST2      |
| 8. | 01 RAD WR WS NISO |
| 8. | 10 RAD WR WS NISO |
| 8. | 11 RAD WR WS NISO |



# PULSE SEQUENCES

## MP0

|     |    |                       |
|-----|----|-----------------------|
| 2.  |    | RSC WG                |
| 3.  |    | RA WB TSGN            |
| 4.  | 0X | RB WL                 |
| 4.  | 1X | RC WL                 |
| 7.  |    | RG WB TSGN2           |
| 8.  |    | RZ WS                 |
| 9.  | 00 | RB WY                 |
| 9.  | 01 | RB WY CI              |
| 9.  | 10 | RC WY CI              |
| 9.  | 11 | RC WY                 |
| 10. |    | RU WB TSGN ST1 NEACON |
| 11. | 0X | WA                    |
| 11. | 1X | WA RB1 RIC 116        |

## MP1

|     |  |             |
|-----|--|-------------|
| 1.  |  | ZIP         |
| 2.  |  | ZAP         |
| 3.  |  | ZIP         |
| 4.  |  | ZAP         |
| 5.  |  | ZIP         |
| 6.  |  | ZAP         |
| 7.  |  | ZIP         |
| 8.  |  | ZAP         |
| 9.  |  | ZIP         |
| 10. |  | ZAP ST1 ST2 |
| 11. |  | ZIP         |

## MP3

|     |    |                   |
|-----|----|-------------------|
| 1.  |    | ZAP               |
| 2.  |    | ZIP NISO          |
| 3.  |    | ZAP               |
| 4.  |    | RSC WG            |
| 5.  |    | RZ WY12 CI        |
| 6.  |    | RU WZ TL15 NFACOF |
| 7.  | 1X | RB WY A2X         |
| 8.  |    | RAD WB WS         |
| 9.  |    | RA                |
| 10. |    | RL                |
| 11. | 1X | RU WA             |

## STD2

|    |  |             |
|----|--|-------------|
| 1. |  | RZ WY12 CI  |
| 2. |  | RSC WG NISO |
| 6. |  | RU WZ       |
| 8. |  | RAD WB WS   |

## PULSF SEQUENCES

### PINC

|    |                     |
|----|---------------------|
| 1. | RSCT WS             |
| 2. | RSC WG              |
| 5. | RG WY TSGN TMZ TPZG |
| 6. | PONFX               |
| 7. | RU WSC WG WQVR      |
| 8. | RB WS               |

### PCDU

|    |                     |
|----|---------------------|
| 1. | RSCT WS             |
| 2. | RSC WG              |
| 5. | RG WY TSGN TMZ TPZG |
| 6. | CI                  |
| 7. | RUS WSC WG WQVR     |
| 8. | RB WS               |

### MINC

|    |                     |
|----|---------------------|
| 1. | RSCT WS             |
| 2. | RSC WG              |
| 5. | RG WY TSGN TMZ TPZG |
| 6. | MONEX               |
| 7. | RU WSC WG WQVR      |
| 8. | RB WS               |

### MCDU

|    |                     |
|----|---------------------|
| 1. | RSCT WS             |
| 2. | RSC WG              |
| 5. | RG WY TSGN TMZ TPZG |
| 6. | MONEX CI            |
| 7. | RUS WSC WG WQVR     |
| 8. | RB WS               |

### DINC

|    |                     |
|----|---------------------|
| 1. | RSCT WS             |
| 2. | RSC WG              |
| 5. | RG WY TSGN TMZ TPZG |
| 6. | 00 MONEX POUT       |
| 6. | 10 PONEX MOUT       |
| 6. | X1 ZOUT             |
| 7. | RU WSC WG WQVR      |
| 8. | RB WS               |

## PULSF SEQUENCES

### SHINC

|    |                 |
|----|-----------------|
| 1. | RSCT WS         |
| 2. | RSC WG          |
| 5. | RG WYD TSGN     |
| 7. | RUS WSC WG WQVR |
| 8. | RB WS           |

### SHANC

|    |                 |
|----|-----------------|
| 1. | RSCT WS         |
| 2. | RSC WG          |
| 5. | RG WYD TSGN CI  |
| 7. | RUS WSC WG WQVR |
| 8. | RB WS           |

### INOTRD

|    |        |
|----|--------|
| 1. | WS     |
| 2. | RSC WG |
| 5. | RCH    |
| 8. | RB WS  |

### INOTLD

|    |        |
|----|--------|
| 1. | WS     |
| 2. | RSC WG |
| 5. | RCH    |
| 7. | WCH    |
| 8. | RB WS  |

## PULSE SEQUENCES

### FETCH0

|    |               |
|----|---------------|
| 1. | RA WS         |
| 2. | RSC WG WY ST1 |
| 4. | WSC           |
| 8. | WS            |

### FETCH1

|     |             |
|-----|-------------|
| 2.  | RSC WG      |
| 7.  | RG          |
| 8.  | RA WS U2BBK |
| 10. | RRBK        |

### STORF0

|    |               |
|----|---------------|
| 1. | RA WS         |
| 2. | RSC WG WY ST1 |
| 4. | WSC           |
| 8. | WS            |

### STORE1

|     |             |
|-----|-------------|
| 2.  | RSC WG      |
| 4.  | WSC         |
| 7.  | RG          |
| 8.  | RA WS U2BBK |
| 9.  | WG          |
| 10. | RRBK        |

51

